

Care of the Older Adult With Chronic Kidney Disease

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CHAPTER OUTLINE

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KEY POINTS

- Changes in muscle mass have a large impact on the most used equations for estimated glomerular filtration rate. This has a greater impact on older adults, particularly in those with functional limitations or clinical frailty, compared with the overall population.
- Older individuals are at higher risk of being misclassified as having chronic kidney disease (CKD). Use of definitions that include age-modified thresholds to define CKD and/or the use of age-dependent formulae that include albuminuria assessments are being proposed.
- Older individuals with CKD appear to have an unexpectedly high burden of geriatric syndromes such as frailty, cognitive impairment, functional limitations, depression, incontinence, sensory deficits, malnutrition, and falls.
- A complete evaluation (known as a comprehensive geriatric assessment) is recommended for all older individuals. This should include a detailed medical history (including treatment targets), assessment of the home environment and social support structure, functional history, and assessment of personal values and lifestyle preferences.
- Patients with CKD are at risk of polypharmacy. Polypharmacy in older adults is further complicated by altered pharmacokinetics and pharmacodynamics in the setting of advancing age and presence of CKD.
- Growing evidence indicates that dialysis does not alleviate the signs and symptoms of advanced kidney disease in older adults.
- Most studies have shown that dialysis in older adults is associated only with a modest survival advantage over comprehensive conservative kidney care.
- Several studies have suggested that dialysis initiation may exacerbate or accelerate functional and cognitive decline, resulting in a 1.6-fold higher risk of serious fall injuries and a higher risk of requiring professional nursing care.
- It is important to distinguish between younger and older individuals, robust or frail, and those with complex clustering of multiple chronic diseases while engaging in shared decision-making processes.

Although chronic kidney disease (CKD) is a significant public health issue for patients of all ages, there is a limited body of literature addressing the management of CKD in older adults. Most health care practitioners would agree, however, that caring for older adults with CKD presents

unique challenges because of the substantial differences between older and younger patients with this condition. In this chapter, we provide an overview of a broad number of issues related to providing care for older adults with CKD. We have included discussions about the debate

rejecting current definitions used to define CKD, detailed information about common geriatric syndromes (e.g., frailty, functional decline, cognitive impairment), and their relationship with kidney disease and treatment. We also introduce the role of palliative dialysis and comprehensive conservative renal care.

CHRONIC KIDNEY DISEASE AND AGE-RELATED CHANGES IN ESTIMATED GLOMERULAR FILTRATION RATE

EVALUATION OF KIDNEY FUNCTION IN OLDER ADULTS

Several studies have suggested that GFR declines steadily with healthy aging. This decline is often at a slow rate in the absence of albuminuria. Although normal aging is said to cause fibrotic changes in the kidney, there is insufficient evidence to suggest that pathologic findings parallel changes in kidney function. In an evaluation of 1203 healthy potential kidney donors, Rule and colleagues¹ correlated the degree of kidney fibrosis on histology with GFR, measured using iothalamate clearance, across a number of age groups. This reduction in kidney function appeared to begin in the third decade but was most discernible by the sixth or seventh decade.² Fibrosis was more common with advancing age, and the measured GFR (mGFR) fell with age (estimated at 4.6 mL/min/1.73 m² per decade in men and 7.1 mL/min/1.73 m² per decade in women). There was substantial heterogeneity in mGFR values among patients with similar degrees of fibrosis and of the same age, and the resulting correlation between mGFR and fibrosis was weak.³ (See [Chapter 21](#) for further discussion on the physiology and pathophysiology of the aging kidney).

Estimates of the prevalence of CKD vary widely from study to study and appear to be highly dependent on the method used to evaluate kidney function and on the population under study.^{4,5} Several observational studies report a high CKD burden among older individuals. In these studies, authors use traditional eGFR thresholds to define CKD (eGFR <60 mL/min/1.73 m² that persists longer than 3 months). Using this definition, CKD prevalence in those aged 65 years and older is approximately 10-fold higher than that seen in those younger than 50 years (40% vs. 4%, respectively).⁶⁻⁸ However, recent studies within living kidney donor populations have characterized age-related decline in kidney function seen in the absence of disease.⁹ On the basis of these data, it seems likely that CKD prevalence is overestimated in the older age group and increasingly the use of a single GFR threshold across all age groups is being contested.¹⁰

Current KDIGO guidelines¹¹ recommend assessment of kidney function using estimating equations (eGFR) in preference to formal assessments. Formal evaluation of kidney function, using inulin or iothalamate, is both impractical and costly in the clinical setting. Similarly, for older individuals, 24-hour urine collections can be cumbersome to collect and are more likely to be unreliable due to inappropriate collection, incontinence, or accidental spills. The widespread use of estimating equations to categorize kidney function is, however, also imperfect in

older, frail populations and may contribute to mislabeling of those both with and without kidney “disease.” Current standards require estimating equations to fall within a 30% margin of the measured GFR in ≥90% of estimates. Thus even at baseline, eGFR estimates incorporate a relatively large error margin into the reported eGFR. This error margin is further amplified in older individuals because of non-GFR determinants resulting in a larger risk of both underestimation and overestimation of kidney function. Non-GFR determinants include, among other factors, a low dietary intake of protein and reduced muscle mass (both factors specific to creatinine levels), subclinical inflammation, and a change in body habitus with increased adiposity. Even in the absence of overt disease, older individuals are more likely to be frail or have chronic inflammation arising from normal age-related conditions (sometimes referred to as “inflammaging”). A number of different equations have been proposed to improve detection of kidney disease within an older population ([Table 60.1](#)), but no one equation appears to be optimal.¹² Consequently, the inaccuracies around eGFR estimation and the observation that older patients have less albuminuria and slower deterioration rates continue to fuel debate as to whether it is appropriate to label individuals, in their eighth and ninth decades, as having a chronic disease based solely on laboratory criteria.¹³

U.S. guidelines continue to recommend use of the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine-based equation¹⁴ and advise additional confirmatory testing with cystatin C or a clearance measurement in specific circumstances. European guidelines specific to the older population¹⁵ allow more flexibility and do not recommend one equation over the other. No guideline, at present, distinguishes among different age groups or among robust, vulnerable, and frail individuals. Revisions to U.S. guidelines are anticipated because most laboratories no longer include race in the computation. However, it is unclear if newer guidelines will also include age-related modifications as data about the correlation to clinically important outcomes including identification of those at risk of accelerated cardiovascular morbidity, progression to end-stage kidney failure (ESKF), and/or identification of those at increased risk of death emerge.¹⁶

PROGNOSTIC AND CLINICAL VALUE OF ESTIMATING KIDNEY FUNCTION

Contemporary definitions of CKD stage 1 to 5 do not differentiate between older and younger adults and likely result in numerous older individuals being mislabeled and potentially overtreated for CKD.¹⁷⁻²⁰

In a U.S.-based observational study of 4562 patients older than 65 years of age, Arora and colleagues²¹ found that 62% of patients with CKD stage 3b and 69% of patients with CKD stage 4 progressed at an annual rate of <1 mL/min/1.73 m². In the Berlin Initiative Study, individuals aged 70 years or more living in Berlin were followed prospectively for a median of 6 years. They found eGFR decline, in men for example, fell from a median of 3 mL/min/year at age 70 years to 0.75 mL/min/year by age 92 years. Declines in renal function appeared to be nonlinear with slower decline

Table 60.1 Estimated Glomerular Filtration Rate (eGFR) Equations

Research Group	Endogenous Filtration Marker (Units)	Equation	Development Population	Limitations
Cockcroft & Gault, 1976 ³⁰⁶	Creatinine (mg/dL)	$\text{CrCl} = ([140 - \text{age}] \times \text{weight}) / [72 \times S_{\text{Cr}}] \times 0.85$ (if female)	$N = 505$ (236 for training sample) whites, 4% women hospitalized pts Age range: 18–92 yr (only 17 patients > 80 yr)	Values not adjusted for BSA Developed to predict CrCl instead of GFR Underestimates GFR in older adults, especially at higher GFRs
MDRD, 1999 ³⁰⁷	Creatinine ($\mu\text{mol/L}$)	$\text{eGFR} = 175 \times (S_{\text{Cr}})^{-1.154} \times (\text{age})^{-0.203} \times (0.742 \text{ if female}) \times (1.212 \text{ if black})$	$N = 1628$ whites and blacks; 40% women Nondiabetic pts with CKD Mean age: 51 ± 13 yr (age range, 18–70 yr)	Tends to underestimate kidney function in those with a normal eGFR Not validated in patients >70 yr, at extremes of body weight, and in some ethnic groups (e.g., Asian ethnic groups)
CKD-EPI, 2009, ¹⁴ 2012 ³⁰⁸	CKD-EPIcr—creatinine (mg/dL)	$\text{eGFR} = 141 \times \min(S_{\text{Cr}/\kappa,1})^{\alpha} \times \max(S_{\text{Cr}/\kappa,1})^{-1.209} \times 0.993^{\text{age}}$ ($\times 1.018$ if female) ($\times 1.159$ if black) where κ is 0.7 for females and 0.9 for males	$N = 8254$ nonblacks and blacks, 44% women Mean age: 47 ± 15 yr (only 216 patients > 70 yr and 41 > 80 yr)	Developed in a population with few patients >70 yr and almost none >80 yr Less accurate in men with $\text{BMI} \geq 35 \text{ kg/m}^2$
	CKD-EPIcys—cystatin C (mg/L)	$\text{eGFR} = 133 \times \min(S_{\text{cys}/0.8,1})^{-0.499} \times \max(S_{\text{cys}/0.8,1})^{-1.328} \times 0.996^{\text{age}}$ [$\times 0.932$ if female] where min indicates the minimum of $S_{\text{Cr}/\kappa}$ or 1 and max indicates the maximum of $S_{\text{Cr}/\kappa}$ or 1	$N = 5352$ nonblacks and blacks; 42% women Mean age: 47 ± 15 yr (719 patients > 65 yr)	
LMR ³⁰⁹	Creatinine ($\mu\text{mol/L}$)	$\text{eGFR} = e^{X - 0.0158 \times \text{age} + 0.438 \times \ln(\text{age})}$ Female: $\text{pCr} < 150 \mu\text{mol/L}$: $X = 2.50 + 0.0121 \times (150 - \text{pCr})$ Female: $\text{pCr} \geq 150 \mu\text{mol/L}$: $X = 2.50 - 0.926 \times \ln(\text{pCr}/150)$ Male: $\text{pCr} < 180 \mu\text{mol/L}$: $X = 2.56 + 0.00968 \times (180 - \text{pCr})$ Male: $\text{pCr} \geq 180 \mu\text{mol/L}$: $X = 2.56 - 0.926 \times \ln(\text{pCr}/180)$	$N = 3495$ Swedish whites; 47% women Mean age: 63 yr (age range, 21–86 yr; 756 pts 70–79 yr and 342 patients ≥ 80 yr)	Difficult to apply Overestimation in patients with $\text{BMI} < 20 \text{ kg/m}^2$, especially males Not validated in other racial and ethnic groups (e.g., black, Asian)
FAS, 2017 ³¹⁰ , 2016 ³¹¹	Creatinine (mg/dL)	$\text{eGFR} = \frac{107.3}{(S_{\text{Cr}}/Q_{\text{crea}})} \times 0.988^{(\text{age} - 40)}$ Equation for age >40 yr, where $Q_{\text{crea}} = 0.7 \text{ mg/dL}$ for females and 0.9 mg/dL for males	$N = 6870$ nonblacks; 47% women Mean age: 74 yr (1764 patients ≥ 70 yr).	Lack of external validation May perform better in healthy and general population than in the CKD population Needs further external validation (e.g., blacks and Asians)
	Cystatin C (mg/L)	$\text{eGFR} = \frac{107.3}{(S_{\text{cysC}}/Q_{\text{cysC}})} \times 0.988^{(\text{age} - 40)}$ where $Q_{\text{cysC}} = 0.82 \text{ mg/L}$ when age 40–70 yr and 0.95 mg/L when age ≥ 70 yr	$N = 6132$ nonblacks; 7% women Mean age: 58 ± 18 yr (1469 patients ≥ 70 yr)	Clinical utility needs to be confirmed in further studies
BIS, 2012 ³¹²	BIS1—creatinine (mg/dL)	$\text{BIS1: } 3736 \times \text{creatinine}^{-0.87} \times \text{age}^{-0.95} \times 0.82$ (if female)	$N = 570$ (for equation development; whole cohort, $N = 2073$)	Needs further external validation (e.g., blacks and Asians)
	BIS2—cystatin C (mg/L)	$\text{BIS2: } 767 \times \text{cystatin C}^{-0.61} \times \text{creatinine}^{-0.40} \times \text{age}^{-0.57} \times 0.87$ (if female)	White German participants; 43% women	

BIS, Berlin Initiative Study equation; BMI, body mass index; BSA, body surface area; CKD-EPI, Chronic Kidney Disease–Epidemiology Collaboration equation; FAS, Full Age Spectrum equation; LMR, revised Lund-Malmö equation; MDRD, Modification of Diet in Renal Disease Study equation.

in those reaching the eighth and ninth decades of life.^{16,22} Similar findings have been reported elsewhere^{23,24} and likely reflect the fact that relatively few older patients with an eGFR <45 mL/min have albuminuria.^{25,26} Consequently, inclusion of information about albuminuria is important when interpreting results showing reduced eGFR in older subjects.²⁷ Albuminuria rates are low among those aged older than 70 years. In the Berlin Initiative Study, the albuminuria rate among 2069 adults 70 years of age and older was 26%. Unlike younger individuals, only a small percentage had macroalbuminuria (3.6% had an albumin-to-creatinine ratio [ACR] ≥ 300 mg/g).²⁸ Similar results were found in the National Health and Nutrition Examination Survey (NHANES) III population, where 21% presented with albuminuria at 70 years of age and older and 32.7% at 80 years and older.²⁹

The lower rate of kidney decline observed in older individuals, especially those older than 75 years, means that older adults are more likely to die than progress to ESKF, even when eGFR is severely reduced. In a large Veterans Administration cohort, the risk of death exceeded that of initiating dialysis in adults older than 85 years, regardless of eGFR (Fig. 60.1).²³ In a community-dwelling population of 1268 older individuals (mean age, 75 years) with an eGFR <60 mL/min/1.73 m² followed for 9.7 years, death was 13 times more likely than progression to ESKF, and 25 times more likely for those between 76 and 85 years of age. Even among those who had more advanced CKD (eGFR <45 mL/min/1.73 m²), the risk of death remained more than sixfold higher than the risk of ESKF.³⁰

Notwithstanding the mortality data, there continues to be an association between the presence of CKD and increased vascular events extending across all age groups.³¹⁻³⁴ In the Cardiovascular Health Study, a community-based cohort of U.S. adults aged 65 years and older, individuals with CKD had more than twice the prevalence of coronary artery disease compared with those without CKD.^{35,36} In three large cohorts, the Kidney Early Evaluation Program (KEEP; $N = 27,017$), NHANES, 1999 to 2006 ($N = 5538$), and the

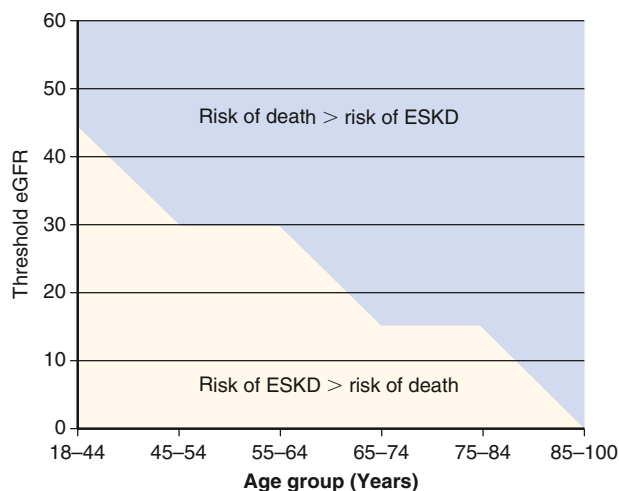


Fig. 60.1 Baseline estimated glomerular filtration rate (eGFR) threshold below which the risk for end-stage kidney failure (ESKF) exceeded the risk of death for each age group. (From O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol*. 2007;18:2758-2765.)

Medicare 5% sample ($N = 1,236,946$), the prevalence of diabetes in those with CKD ranged from 21.4% to 46.2% and that of hypertension was 90% or higher.³⁴ High rates of concomitant high cholesterol, coronary artery disease, congestive heart failure, cardiovascular disease, and cancer were also seen in the KEEP and NHANES cohorts.

ROLE OF CLINICAL PREDICTION MODELS IN OLDER PERSONS

Clinical prediction models to assess the risk of death specifically among older patients with CKD are rare and not widely used.³⁷⁻³⁹ Instead, prediction models that predict the likelihood that an individual will progress to ESKF have been widely incorporated into clinical practice. Using two large Canadian cohorts of patients with CKD stages 3 to 5 defined using eGFR, Tangri and colleagues^{40,41} developed and subsequently validated in diverse global cohorts ($N = 721,357$), two risk equations, termed the eight- and four-variable kidney function risk equation (KFRE). Addition of other clinical data, such as comorbidity or previous rate of kidney decline, did not significantly improve the predictive power of the KFRE.⁴² The more popular four-variable KFRE uses age, gender, eGFR, and urinary ACR and performs sufficiently well for use in older patients with an eGFR <45 mL/min/1.73 m².¹⁵ One criticism of the four-variable KFRE, particularly in older populations, is that it does not address the risk that an acute illness may be associated with an acute precipitous drop in kidney function. In fact, between 25% and 50% of older individuals starting dialysis do so after an episode of acute kidney injury (AKI).^{43,44} It has been suggested that future modeling equations also include the patient's competing risk of death before dialysis requirements, rate of kidney function decline, and information about the AKI risk.⁴⁵

COMORBIDITIES AND GERIATRIC SYNDROMES AMONG OLDER INDIVIDUALS WITH CHRONIC KIDNEY DISEASE

"Geriatric syndromes" (GSs) is the term used to describe a group of common health conditions in older people that do not fit into discrete disease categories. These conditions include frailty, functional limitation, falls, depression, polypharmacy, malnutrition, and cognitive impairment. Collectively, the GSs arise from a complex interplay among age-related physiologic changes, chronic disease, and functional stressors in older adults. Each interacts with the other, and often the syndromes co-occur.

Emerging data in older individuals with CKD have suggested an unexpectedly high burden of GSs. It is important for nephrology teams to recognize these geriatric issues for several reasons. The appropriateness and effectiveness of medical interventions, such as dialysis care, are dependent on the overall patient well-being, GSs, and general health. Moreover, from the patient's perspective, the presence of one or more GSs almost invariably affects their lifestyle, as well as that of their family and social circle. Timely recognition of these syndromes can help patients and families have more realistic expectations while

participating in treatment decision making and trigger discussions around goals of care and advance care planning (see [Chapter 61](#)).

The formal evaluation of a patient, using a geriatric lens to recognize GSs and incorporate findings into clinical care, is referred to as a comprehensive geriatric assessment (CGA). Ideally, the CGA assessment is conducted over several visits, in the home and in the clinic, by a variety of health care professionals in addition to the physician including nursing, physiotherapy, occupational therapy, and social work. Evaluation includes a detailed medical history, such as treatment targets, assessment of the home environment and social support structure, functional history, and an assessment of personal values and lifestyle preferences. This information is then coupled with the medical issues and assessment of strength, balance, sensory function, cognition, depression, nutrition, and communication skills and used to identify potential areas of concern. Multidisciplinary teams use the results of the assessment to develop a coordinated plan that maximizes health while minimizing the impact of geriatric impairments, or the treatments being initiated, on the individual and his or her caregivers. This multidimensional and multidisciplinary approach to GSs has been shown to be effective in the general population⁴⁶⁻⁴⁸ and across multiple kidney clinical settings.⁴⁹⁻⁵² Awareness of GSs can improve outcomes and well-being,^{49,52} and routine CGA has been recommended by the American Society of Nephrology, Geriatric Advisory Group, for all older CKD patients.⁵³

FRAILITY

In the clinical realm, frailty is a multidimensional syndrome (or state) associated with increased vulnerability. It often results from growing late-life morbidity combining problems occurring across different domains of daily functioning including physical, sensory, psychological, and social. Frailty has multiple causes and contributors and is characterized by a diminished capacity of the physiologic function to appropriately respond to a stressor, whether physical or psychological. Physical frailty increases the risk of dependency and/or death.^{54,55} Operational definitions for frailty and how to measure it vary, and a multitude of validated scales have been developed over the past decades.⁵⁶ This contributes to much of the variability in estimated prevalence. The sole use of the physical domain as a surrogate measure for the global assessment of frailty, such as the classification by Fried and colleagues,⁵⁷ in which definitions are based on five physical components—unintentional weight loss, decreased strength, decreased exercise tolerance, reduced gait speed, and fatigue—may be an inadequate metric for frailty, particularly in CKD. Social determinants of frailty, such as those associated with environmental and mental well-being, are also of importance and may be recognized through more holistic assessments.^{58,59} The Clinical Frailty Scale is an example of a simple and clinically applicable validated tool that can be used to assess global frailty status. The scale uses several grades of frailty ([Fig. 60.2](#)) based largely on the clinical judgment of the physician, making it an easily applied screening instrument to identify those who would most benefit from a formal CGA, which remains the gold standard for frailty assessment.

Frailty is of clinical importance because it is associated with an increased risk of both morbidity (e.g., falls, hospitalization, dependence, need for long-term care) and mortality.⁶⁰⁻⁶² Risks of mortality are doubled in those with frailty compared with nonfrail individuals across the complete spectrum of CKD. For example, in CKD patients enrolled in the MDRD study, physical frailty determined from self-reported questionnaires was associated with a 1.7-fold increase in mortality over time (hazard ratio [HR] 1.7; 95% confidence interval [CI], 1.3–2.3).⁶³ A special U.S. Renal Data System study also found physically frail patients starting dialysis had a 1.8-fold higher mortality over a 3-year period (HR, 1.8; 95% CI, 1.4–2.2).⁶²

The prevalence of frailty appears to increase as kidney function declines^{63,64} and is estimated to be twofold to fourfold higher in the kidney population, compared with those with normal kidney function.⁶⁴⁻⁶⁶ Recent prevalence estimates have suggested that between 40% and 60% of those on dialysis have frailty characteristics.^{60,67,68} Although more prevalent in those with increased age, there is also an unexpectedly high burden of frailty in younger individuals on dialysis.^{69,70} This high prevalence may be partly explained by concomitant CKD-related conditions such as protein-energy wasting (PEW), anemia, inflammation, acidosis, and hormonal disturbances.⁷¹ There is also an increasing awareness of how kidney care treatments may inadvertently promote the sick role, create barriers to routine participation in exercise programs, and decrease participation in other positive health behaviors.⁶⁰

Frailty is a dynamic process in which transitions across states of frailty are common. While transitions can be bidirectional, changes to a worse state are more common than to a better state.⁷²⁻⁷⁴ Although characterized by a decline in physical function, including physical activity and exercise capacity, frailty is often also associated with reduced social participation and self-care abilities, depression, and cognitive impairment and should trigger open discussions reviewing the current goals of care, prognosis, and more frequent evaluation of symptoms. To date, there are no simple one-step interventions that reverse frailty, but multidisciplinary care, nutritional supplementation, exercise, and rehabilitation may attenuate the morbidity associated with frailty, particularly when applied early.^{33,71}

FUNCTIONAL IMPAIRMENT

Functional status describes an individual's ability to perform tasks associated with personal care and maintaining a household. In contrast to leisure activities (e.g., gardening and sports), functional tasks are often regarded as fundamental for daily life, and loss of independence is associated with reduced quality of life (QOL) for patients and their caregivers. Although patients with frailty characteristics have a high burden of functional dependence, both frailty and functional loss may occur independently of each other.

Previous estimates have shown that individuals with CKD stage 3b have a threefold increased risk of developing dependence in daily activities, such as bathing, dressing, and personal care, compared with individuals without kidney impairment.⁷⁵ Although causation cannot be proven, several studies have suggested that dialysis initiation may

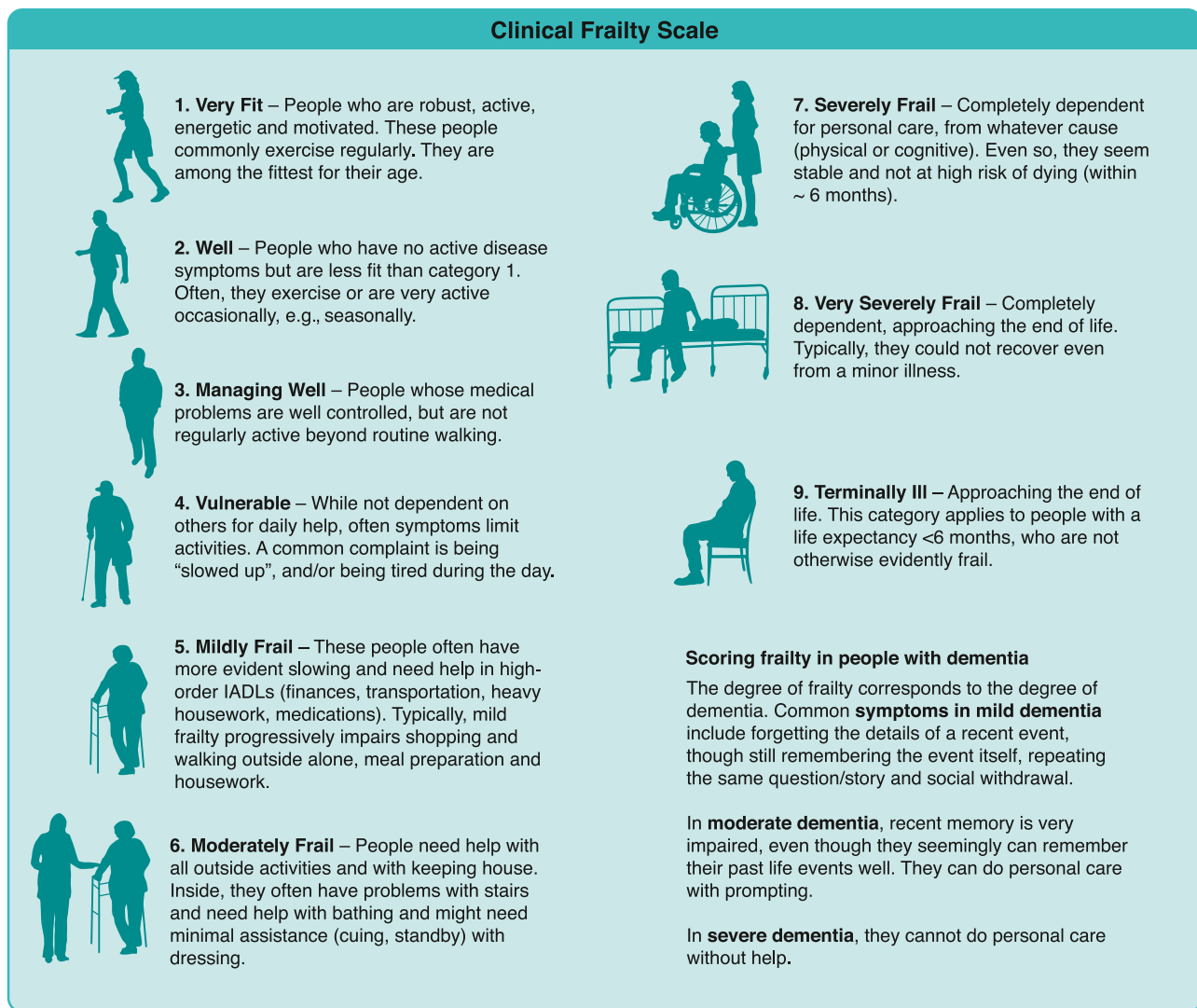


Fig. 60.2 Clinical frailty scale. IADLs, Instrumental activities of daily living. (From Rockwood K, Song X, MacKnight C, et al. A global clinical measure of fitness and frailty in elderly people. *CMAJ*. 2005;173(5):489–495.)

exacerbate or accelerate functional decline (Fig. 60.3) and increase caregiver burden.^{76–80} Most at risk are those with multiple comorbidities, those of older age, and those with functional loss at the time of dialysis initiation. In the Dialysis Outcomes and Practice Patterns Study (DOPPS), 77% of noninstitutionalized patients on maintenance hemodialysis (HD) aged 70 years or older reported dependency,⁸¹ in contrast to 15% of older adults without CKD (≥65 years) in the Cardiovascular Health Study.⁸² Similar levels of functional dependence are seen in those maintained on both HD and peritoneal dialysis.^{81,83}

In the general geriatric literature, functional dependence is recognized as a contributor to subsequent disability, recurrent hospitalization, and increased mortality.^{84–89} Several studies have shown an association between poor mobility status (i.e., requiring assistance with or total dependency for transfers and ambulation) and mortality risk among incident dialysis patients (Table 60.2)^{90–96} and predialysis patients.⁹⁷ Functional dependency (increased need for nursing care, less participation in social and societal roles) is associated with a twofold increase in mortality (HR,

2.4; 95% CI, 1.9–2.9) for those with the highest degree of functional loss.⁸¹

Many factors contribute to the high rate of functional dependencies among older patients with advanced CKD. High rates of depression, cognitive impairment, poor physical functioning, multimorbidity, and the need for repeated hospitalizations may all contribute to the patient's reduced overall health and self-care ability. Early identification of functional impairment in CKD patients may provide an opportunity to ensure adequate management of modifiable contributing factors, provide support and referral to geriatric rehabilitation services, and trigger discussions, with the family and patient, around the prognosis and identification of the trajectory of functional decline in older patients starting dialysis with limitations.

FALLS

As noted previously, several GSs tend to occur concomitantly. Accidental falls are more common in individuals with other GSs, such as concomitant frailty and functional

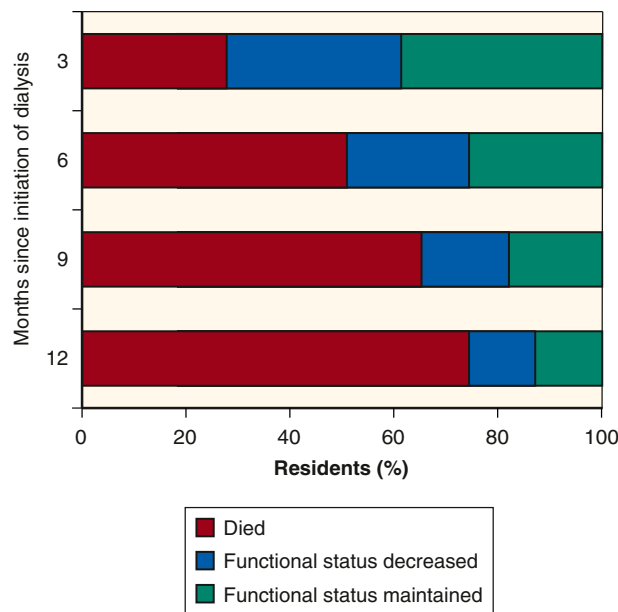


Fig. 60.3 Survival and functional status after the initiation of dialysis among nursing home residents. (From Kurella Tamura M, Covinsky KE, Chertow GM, et al. Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med*. 2009;361:1539.)

decline, although this relationship is not absolute.⁹⁸ Falls are common among older adults and are a leading cause of injury-related hospitalizations, often with prolonged hospital stays, functional decline, long-term care admission, and mortality.⁹⁹⁻¹⁰³ Up to one-third of community-dwelling individuals older than 65 years fall annually, often recurrently.¹⁰⁴ Injurious falls, such as those causing fractures, head injuries, or cuts, occur in one out of five falls,^{105,106} although usually one fall will trigger a cycle of fear of falling (>25% of cases) and a reduction in activity levels, furthering the risk of subsequent falls and the ability to care for oneself.¹⁰⁷

Falls are almost twofold more common in the dialysis population, in both those maintained on HD and those on peritoneal dialysis (PD).^{108,109} Although there are multiple common risk factors for falls (Table 60.3), one of the less appreciated complications of HD that may contribute to falls is the effect of forced immobility. The long time during which patients are sitting relatively still in the dialysis chair may lead to changes in balance, lack of sensory stimulation, and, over time, muscle atrophy. More concerning, however, is the 1.6-fold higher risk of serious fall injuries (i.e., resulting in fractures, head injury, or joint dislocation) that occur in the year after dialysis initiation, compared with the year before dialysis care.¹¹⁰ Clinical data also support the suggestion that dialysis itself may be causative. In a small single-center study of community-dwelling seniors, falls were more common on dialysis days than on nondialysis days (1.45 vs. 1.35 falls/person, respectively) and, if occurring on dialysis days, threefold more common after dialysis than before dialysis.¹⁰³ The association between falls and earlier stages of kidney disease has been less well studied, but some authors have suggested that gait abnormalities, such as shorter stride length and greater time in the stance and double-support phases, are more common with a reduction in eGFR and with acidosis.¹¹¹

Routine clinical questioning about previous falls and conducting simple tests for gait and balance may be a key strategy to identify at-risk individuals who warrant a more detailed evaluation and referral for a CGA. Multidisciplinary risk assessment, followed by tailored multifactorial interventions, such as exercise, treatment of pain, assistive devices, medication review, and home hazard modifications, may be effective in reducing fall risk.

DEPRESSION

Depressive symptoms and clinical depression are common in patients across the complete spectrum of CKD. The prevalence of depression varies, depending on the assessment tools used and the population studied, but, using self-report or screening questionnaires, it is estimated that close to 40% of those undergoing maintenance dialysis have some depressive symptomatology, even if they do not fulfill all criteria for a major depressive episode. Prevalence estimates of major depressive episodes, using formal psychiatrist interview assessments, are lower, ranging from 21% in earlier stages of CKD to 23% in those undergoing maintenance dialysis.¹¹² These estimates may, however, be influenced by the observation, from a number of studies, that patients on dialysis are often unwilling to undergo formal psychiatric evaluation or treatment. Age-specific data are not available, but depression is often more common in socially isolated individuals and those with functional or cognitive impairment placing frail older individuals at increased risk. In addition, often co-occurring with other GSs, depression can be associated with a high burden of pain, fatigue, poor sleep, pruritus, and nausea.¹¹³

The recognition of depression in the older individuals is important for several reasons. In addition to the well-recognized impact on QOL,¹¹⁴⁻¹¹⁷ individuals who are maintained on dialysis and are older than 75 years are among those at highest risk of suicide.¹¹⁸ Furthermore, depression is associated with increased hospitalization and mortality in both CKD and ESKF.¹¹⁹⁻¹²⁴

The approach to treating depression in older patients is the same as for younger patients. The mainstay treatments are depression-specific psychotherapies, such as cognitive behavioral therapy or interpersonal therapy, and pharmacologic treatment with an antidepressant.

Several studies have shown cognitive behavioral therapy to be successful in patients with ESKF¹²⁵⁻¹²⁷ and, where possible, it should be considered the first-line treatment for patients with mild to moderate depression.¹²⁸ However, little is known about the influence of physical diseases, frailty, and cognitive impairment on the efficacy or feasibility of psychotherapy. Despite the high prevalence and clinical implications of depression, many patients remain untreated. Several studies have shown low treatment acceptance and, on average, only one third of patients who may benefit actually receive antidepressant medication.¹²⁹⁻¹³³

Pharmacologic therapies have been used in patients with CKD or ESKF; however, there is little information on safety, efficacy, and optimal dosing.^{134,135} Most antidepressant medications are highly protein bound and are not removed by dialysis. Although they commonly undergo hepatic metabolism, the active metabolites are excreted via the kidney, and consequently dosing should be adjusted to the

Table 60.2 Association of Functional Impairment at Time of Dialysis Initiation with Adverse Health Outcomes

Study, Date of Study	Patients	Study Setting	Functional Assessment	Outcome	Association
Arai et al., ⁹¹ 2014	N = 202 Mean age, 80 yr	Single-center study, incident to dialysis Age ≥75 yr	Mobility (ability or lack of ability to walk without assistance)	6-mo mortality	Higher mortality if unable to walk without assistance; 6-mo mortality risk: aHR, 4.9; 95% CI, 1.4-17.1
Couchoud et al., ⁹² 2009	N = 2500 Mean age, 81 yr	French REIN Registry data, incident to dialysis Age ≥75 yr	Mobility (walk without help, assistance with or total dependency for transfers)	6-mo mortality	Higher mortality if fully dependent for transfers; 6-mo mortality risk: aOR, 1.7; 95% CI, 1.4-2.0
Couchoud et al., ⁹³ 2015	N = 12,500 Age (median), 81 yr	French REIN Registry data, incident to dialysis Age ≥75 yr	Mobility (walk without help, assistance with or total dependency for transfers)	3-mo mortality	Higher mortality if requiring assistance for transfers; 3-mo mortality risk: aOR, 2.5; 95% CI, 2.1-2.9 Higher mortality if totally dependent for transfers; 3-mo mortality risk: aOR, 6.5; 95% CI, 5.4-7.9
Dusseux et al., ⁹⁰ 2015	N = 8955 Age (median), 78 yr	French REIN Registry data, incident to dialysis Age ≥70 yr	Mobility (walk without help, assistance with or total dependency for transfers)	3-yr mortality	Higher mortality if requiring assistance for transfers; 3-yr mortality risk: aOR, 1.7; 95% CI, 1.5-1.9 Higher mortality if total dependency for transfers; 3-yr mortality risk: aOR, 3.0; 95% CI, 2.3-3.8
Glaudet et al., ⁹⁴ 2013	N = 557 Mean age, 71 yr	French REIN Registry data, incident to dialysis	Mobility (walk without help, assistance with or total dependency for transfers)	4-yr mortality	Higher mortality if requiring assistance for transfers; 4-yr mortality risk: aHR, 1.4; 95% CI, 1.2-1.7 Higher mortality if total dependency for transfers; 4-yr mortality risk: aHR, 1.4; 95% CI, 1.0-2.0
Kurella et al., ⁹⁶ 2007	N = 83,996 Mean age, 84 yr	USRDS registry data, incident to dialysis Age ≥80 yr	Mobility (inability to walk or transfer)	Mortality	Higher mortality if unable to walk or transfer Overall mortality risk: aRR, 1.5; 95% CI, 1.5-.6
Thamer et al., ⁹⁵ 2015	N = 52,796 Mean age, 77 yr	USRDS registry data, incident to dialysis Age ≥67 yr	Assistance with daily living (requiring assistance with daily living, inability to ambulate, or has an amputation)	3-mo mortality	Higher mortality if requiring assistance with daily living or for walking; 3-mo mortality risk: aOR, 1.4; 95% CI, 1.3-1.5

aHR, Adjusted hazard ratio; aOR, adjusted odds ratio; aRR, adjusted relative risk; USRDS, U.S. Renal Data System.

eGFR level (Table 60.4). On the basis of current data, selective serotonin reuptake inhibitors (SSRIs), such as sertraline, escitalopram, and citalopram, should be considered first-line for pharmacologic treatment for depression in patients with kidney disease.^{136,137} Tricyclic antidepressants and monoamine oxidase inhibitors (MAOIs) should be avoided in older patients with (or without) kidney disease because of cardiac and central nervous system side effects, including prolonged QTc, arrhythmia, anticholinergic effects, and orthostatic hypotension. Paroxetine may be useful when treating a depressed patient with concomitant intractable pruritus because of its antipruritic actions; drugs such as duloxetine may be prescribed for those with concomitant pain.¹³⁸ Treatment doses of duloxetine need to be adjusted and, based on data from a small pharmacokinetic study

in HD patients,¹³⁹ ideally given at 48-hour intervals. It is important to note that in many regions of the world, this is an off-label prescription, with current U.S. product labeling recommending that the drug not be administered if creatinine clearance (CrCl) is <30 mL/min. Mirtazapine, a noradrenergic and specific serotonergic antidepressant medication, has hypnotic, appetite stimulant, and antipruritic effects and can be used at reduced doses in older kidney patients with concomitant anorexia, sleep disturbance, and/or pruritus.

POLYPHARMACY

Polypharmacy is often defined as a medication count of five or more medications.^{140,141} Because patients with CKD take

Table 60.3 Common Risk Factors for Falls in Older Patients with Chronic Kidney Disease

Risks	Details	References
Previous falls	One or more falls during the previous 12 mo	100,103,313
Fear of falling	Fearful anticipation of a fall; detectable on screening using the Falls Efficacy Scale (high scores)	107
Frailty	Predisposed to falls for multiple reasons (see text)	109,314
Balance and gait problems	Increased postural sway, failed walking test; can be exacerbated by arthritic changes or neuropathy	111,315,316
Pain	Pain may lead to guarding and abnormal posture due particularly to lower limb and foot pain	
Depression and antidepressants	Fall risk worsened by insomnia and sleep fragmentation, impaired attention and psychomotor slowing, and/or exacerbated using sedatives and antidepressants	313,315
Number of prescribed oral drugs	Increased fall risk, particularly with benzodiazepines, antidepressants, β -blockers, other antihypertensive agents, opiate derivatives	315,317,318
Cognitive impairment	Falls precipitated by decreased awareness of environmental risks; altered judgment and reduced executive functioning result in less effective compensation for age- or disease-associated changes in gait and balance	316,319
Sensory deficits	Poor vision and/or hearing affect environmental awareness, balance, and compensation strategies	
Diabetes	Impaired vision due to retinopathy, peripheral neuropathy, and hypoglycemic episodes increases the risk of falls	315,320
Anemia	May result in low energy, poor exercise tolerance, and postural hypotension	
Protein-energy wasting and sarcopenia	Loss of muscle mass and strength results in less effective strategies to compensate and prevent falling; reduced exercise capacity	313,321
HD therapy	Increased risk of transient or rapid fluid and electrolyte shifts, hypotension, arrhythmias; stiffness and muscle weakness exacerbated by forced mobility	322
PD therapy	Environmental risks—connected to a cyclor machine overnight, tubing around the bed may pose trip risk; mechanical factors—changes in the center of gravity due to large volumes of intraabdominal fluid	108

on average 8 medications, and those on dialysis take between 10 and 12 medications, polypharmacy is possibly one of the largest issues in kidney medicine. In older populations, polypharmacy is associated with increased mortality, adverse drug events and drug interactions, falls and other GSs, low medication adherence, and greater health care costs.¹⁴² Safe medication prescribing in older adults with CKD is complicated by not only the number of medications and presence of comorbidities but also the altered pharmacokinetics and pharmacodynamics in the setting of advancing age and presence of CKD. Pharmacist-led deprescribing interventions are effective and recommended.¹⁴³ The combination of age-related changes and CKD places older patients at increased risk of drug accumulation and higher potential for adverse drug effects.^{144,145} Evidence to guide prescribing is severely limited because a large proportion of clinical trials excluded those older than 65 years, and therefore treatment decisions were often based on evidence extrapolated from other patient groups.¹⁴⁶⁻¹⁴⁸ As a general guide, clinicians are advised to consider multiple factors listed in Table 60.5. Drugs that have a narrow therapeutic index and/or high toxicity levels should only be used when other alternatives are not available, and individuals should be monitored carefully.¹⁴⁹ Dialysis patients prescribed opioids, benzodiazepine, gabapentinoids, and baclofen are at risk of dose-dependent increased risks and caution is strongly recommended (see Chapter 61).¹⁵⁰⁻¹⁵²

Several tools can identify medications in which the risks of use in older adults often outweigh the benefits or identify medications that have been omitted but are likely to

benefit the individual. The most widely used tools include The American Geriatrics Society Beers Criteria¹⁵³ and the STOPP/START criteria for Potentially Inappropriate Prescribing in Older People.¹⁵⁴ The 2023 AGS Beers Criteria features a list of medications to avoid or adjust dose in older adults, specifically for those with reduced kidney function. Of importance, they recommend increased monitoring for older adults starting SGLT2 inhibitors because of the clinical impact of urogenital infections and ketoacidosis.

MALNUTRITION AND PROTEIN-ENERGY WASTING

Depending on the definitions used, anywhere between 25% and 75% of older patients maintained on dialysis have evidence of uremic malnutrition, muscle wasting, and body fat loss.¹⁵⁵ In the kidney literature, the term “protein-energy wasting” (PEW) is used in preference to malnutrition because CKD-related factors contribute to the development of wasting, independent of the nutrient intake.¹⁵⁶ A full review of PEW is beyond the scope of this chapter (see Chapter 55); however, it is important to recognize PEW in the older patients because it is associated with weakness, falls, and increased mortality and remains largely unrecognized and undertreated.^{157,158}

Appetite loss is relatively common in older patients with ESKF, with an estimated one-third to one-half of patients experiencing a progressive, spontaneous decrease in food intake as CKD progresses.^{159,160} This change in appetite often drives the initial changes in nutrient intake¹⁵⁶; however,

Table 60.4 Safety, Dosing, and Efficacy of Common Antidepressant Medications used in Chronic Kidney Disease (CKD) and End-Stage Kidney Disease

Drug Class, Generic	Dose Adjustment in CKD ¹³⁴	Comment	Efficacy Studies	Class Adverse Effects ^{134,135,323}
Selective serotonin reuptake inhibitor (SSRI)				
Sertraline	CKD stages 1-4: no dosage adjustment required; 50-200 mg/day CKD stages 5-5D: start at 25 mg/day, consider decreasing max dose	<1% excreted unchanged in urine Pharmacokinetics in CKD are unchanged in single-dose studies, but no published data on multiple dosing Likely not dialyzed	Three prospective efficacy studies ^{131,324,325} ; sertraline significantly improved BDI scores	Common: insomnia, restlessness, nausea, headache, GI upset, sexual dysfunction, activation (mainly with fluoxetine and sertraline)
Paroxetine	IR: 10-40 mg/day CR: 12.5-50 mg/day	<2% excreted in urine Increased plasma concentration found when GFR <30 mL/min If GFR <30 mL/min, start at 10-20 mg/day, increase slowly Likely not dialyzed, decrease total daily dose by 50%	One study among HD patients ($N = 34$) ³²⁶ ; paroxetine significantly improved depressive symptoms and nutritional markers	Rare: SIADH, increased bleeding risk, extrapyramidal symptoms, serotonin syndrome (in combination with other serotonergic drugs), and QT prolongation (seen with dose >40 mg of citalopram)
Citalopram	No dosage adjustment required	<15% excreted in urine Manufacturer does not recommend use if GFR <20 mL/min Dose adjustment normally not required in kidney impairment, but use with caution when GFR <10 mL/min Only 1% of drug removed by HD	One randomized study (citalopram vs. psychological training) in HD population ($N = 44$) ³²⁷ ; both citalopram and psychological training significantly reduced HADS scores at the end of 3 mo	
Escitalopram	No dosage adjustment required	<10% excreted in urine Manufacturer does not recommend use if GFR <20 mL/min Likely not dialyzed	One randomized study ³²⁸ (escitalopram vs. placebo) in 58 ESKFESKF patients—escitalopram significantly improved HADS scores compared with placebo	
Fluoxetine	No dosage adjustment required Use with caution.	5%-10% excreted in urine Long half-life. If GFR < 20 mL/min, consider using on alternate days or low dose Likely not dialyzed	Two small prospective efficacy studies—fluoxetine improved depressive symptoms by more than 25% in 6 HD patients ³²⁹ and, compared with placebo, improved depression at 4 wk but not 8 wk in 14 HD patients ³³⁰	
Serotonin-Norepinephrine Reuptake Inhibitor (SNRI)				

Continued on following page

Table 60.4 Safety, Dosing, and Efficacy of Common Antidepressant Medications used in Chronic Kidney Disease (CKD) and End-Stage Kidney Disease (Continued)

Drug Class, Generic	Dose Adjustment in CKD ¹³⁴	Comment	Efficacy Studies	Class Adverse Effects ^{134,135,323}
Venlafaxine	Normal dosage: 75-225 mg/day eGFR 10-70: consider reducing dose by 25%-50%. Dialysis: reduce total daily dose by 50%	5% excreted in urine, clearance decrease and half-life prolonged in kidney impairment Manufacturer advises to avoid use if GFR <10 mL/min Accumulation of toxic metabolite can occur, rhabdomyolysis and kidney failure have been reported but rare	No efficacy data	Like SSRIs plus increased blood pressure Liver toxicity seen with duloxetine
Duloxetine	No adjustment required if eGFR >30 Recommend starting at 30 mg every other day, titrating slowly to a max 60 mg daily; use not recommended with eGFR <30	Off-label use in patients with advanced CKD and on maintenance dialysis	No efficacy data	
Noradrenergic and Specific Serotonergic Antidepressant				
Mirtazapine	No dosage adjustment recommended for eGFR 10-50 Start at 3.75 mg and up-titrate to effect, max 45 mg daily GFR <10: consider dose reduction	75% excreted unchanged in urine Clearance reduced by 50% in those with GFR <10 mL/min Has been used to treat pruritus caused by kidney failure	No efficacy data	Appetite stimulation, weight gain, sedation
Norepinephrine-Dopamine Reuptake Inhibitor				

Table 60.5 Key Elements in Considering Risks and Benefits when Prescribing for Older Adults with Kidney Disease

Risk Considerations	Benefit Considerations
Medication-Associated risk	
Is the medication cleared in whole or in part by the kidney?	What is the population in which this medication has been studied?
Does the medication have a narrow therapeutic window?	Does it include older adults or those with kidney disease? Is there observational or clinical trial evidence that benefits extend to those with kidney disease?
Is the medication thought to be of high risk in the general older population or in individuals with comorbidities, similar to those of the patient in question?	If patients with kidney disease have not been studied, does the medication have a track record of safety in postapproval studies?
Does the medication have potential central nervous system effects?	Would benefits of the medication be due to improvement in symptoms or in decrease in risk from asymptomatic disease? What are the patient's preferences in adding new medications?
Are data available to guide dosing in patients with kidney disease (e.g., pharmacokinetic studies, drug level monitoring)?	Does the medication address a problem for which the patient is at significant risk (e.g., cardiovascular disease in patients with kidney disease)? Is the patient likely to accrue significant absolute risk reduction from the new medication?
Patient-associated risk	
Is the patient already taking multiple medications?	
Does the patient have cognitive dysfunction, poor vision, frailty?	
Risk of additional medications may be higher in this group.	
Does the patient have a history of adherence problems, and what would be the consequences of patient-related erratic dosing?	

From Rifkin DE, Winkelmayer WC. Medication issues in older individuals with CKD. *Adv Chronic Kidney Dis.* 2010;17:320-328.

the spiral of decline in overall nutrition is compounded by multiple factors common among older individuals (Fig. 60.4). In kidney medicine, the introduction of dietary restrictions affects variety, flavor, and palatability of food and further increases undernutrition. In many cases, the loss of the social pleasures of eating leads to a low QOL, in addition to PEW.

Special attention should be given to nutritional status in older individuals, and personalized support from a kidney dietitian is imperative when PEW is noted—both the subjective global assessment (SGA) and malnutrition inflammation score (MIS) are reliable screening tools.

Interventions to improve PEW are often limited to reducing dietary restrictions and adding oral supplements, with little evidence to support dialytic interventions. Protein intake should not be restricted for elderly patients with CKD because significant weight loss is seen in older people on low-protein diets.¹⁶¹ Appetite stimulants such as mirtazapine can be tried. Constipation and/or dry mouth should be aggressively treated. Where possible, drugs known to reduce appetite, such as calcium binders and semaglutide, should be avoided.

COGNITIVE IMPAIRMENT

Cognitive impairment affects patients negatively by contributing to functional dependence and behavioral symptoms, which in turn have important effects on patient well-being and QOL. Impaired cognition potentially affects multiple areas of patient care including patient adherence to treatment plans and medication and may lead to premature institutionalization and death.¹⁶²⁻¹⁶⁴ Pathophysiology,

prevalence, and treatments of neurologic conditions in kidney disease are discussed elsewhere, but it is important to note that the HD itself is associated with metabolic changes, consistent with transient ischemia, and may contribute to cognitive decline.¹⁶⁵ Screening is important for all individuals, and relatively quick tests such as the Mini-Cog (three-word recall and clock drawing) or the Mini-Mental State Examination may be useful in a busy practice. In the dialysis population, administration of the Montreal Cognitive Assessment, preferably within the first 2 hours of starting dialysis, appears to be the most reliable cognitive screening tool.¹⁶⁶ Because cognitive impairment deteriorates over time, particularly after dialysis initiation,¹⁶⁷ it is important to recognize and discuss any features of cognitive impairment that are present and introduce the patient and family to the high likelihood of further progression to dementia. Implementation of adequate social support and planning for future patient care is a critical part of these discussions.

CONTROVERSIES AND COMPLEXITY IN THE MANAGEMENT OF OLDER PATIENTS WITH CHRONIC KIDNEY DISEASE

Over the past decade, multiple clinical practice guidelines have emerged, creating an evidence-based structure that is currently used to guide clinical CKD care. Clinical practice guidelines help standardize care and ensure a consistent approach, across all patients, that includes the interventions most likely to reduce mortality and cardiovascular events and to slow disease progression. Guidelines incorporate data from large population studies and consequently their

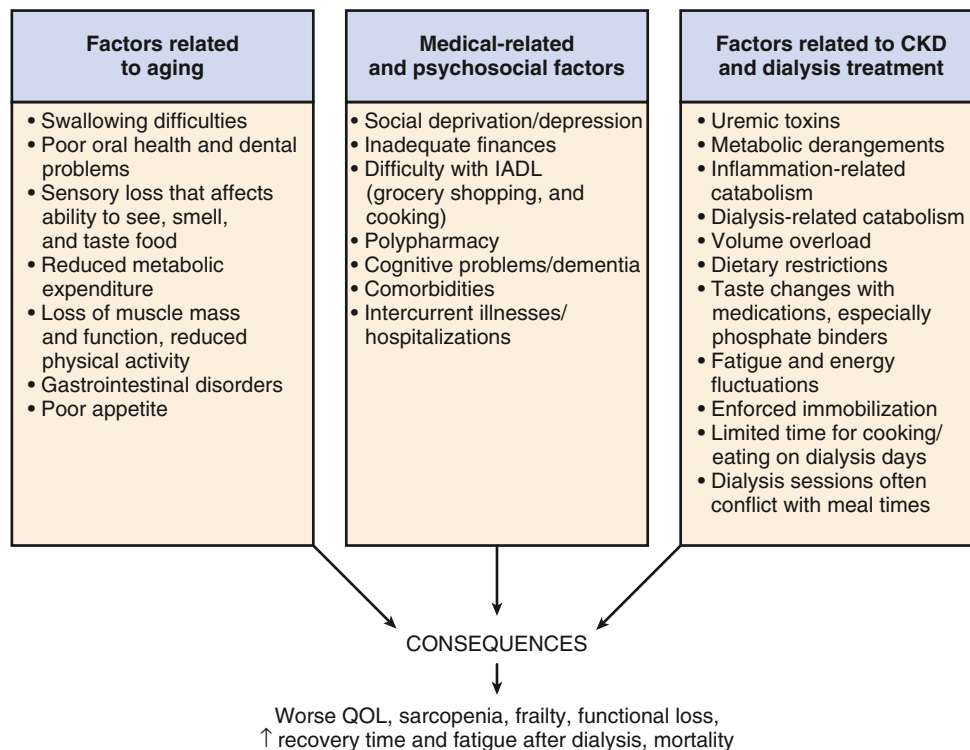


Fig. 60.4 Common factors related to aging and dialysis leading to malnutrition and protein-energy wasting (PEW). CKD, Chronic kidney disease; IADL, instrumental activities of daily living; QOL, quality of life.

strength—the ability to create a simplified model to guide consistent high-quality care—is also, in some respects, their greatest weakness. Individual patients bring their own complexities, values, and goals and, particularly when applied to frail older individuals approaching the end of life, guidelines may lead to care that is poorly aligned with the individual's needs. Clinicians are encouraged to solicit individual patient goals and values; distinguish between younger and older individuals, robust or frail; and distinguish between those with kidney-limited CKD and those with complex clustering of multiple chronic diseases.

Within the kidney sphere, guidelines have given rise to several discussions around whether adaptation is required when treating an older frail patient. These controversial issues are introduced in the subsequent sections.

HYPERTENSION IN THE FRAIL OLDER PATIENT WITH CHRONIC KIDNEY DISEASE: TARGETING BLOOD PRESSURE

Decades of research have unequivocally shown that antihypertensive treatment reduces the risk of cardiovascular (CV) outcomes in several populations including older and very old patients. Currently, however, there is uncertainty about how strict blood pressure (BP) control should be in older individuals with advanced kidney disease.^{168,169} Although a discussion of all hypertensive trials is beyond the scope of this chapter and is well covered elsewhere (see [Chapter 48](#)), data pertinent specifically to older individuals with CKD are included later.

Newer guidelines have incorporated data from the Systolic Blood Pressure Intervention Trial (SPRINT) published in 2015 promoting systolic BP targets <120 mm Hg.¹⁷⁰ SPRINT is a prospective randomized controlled trial, funded by the National Institutes of Health, that has randomized 9361 patients with high CV risk to standard treatment targeting systolic BP <140 mm Hg or to intensive BP-lowering targeting systolic BP <120 mm Hg. SPRINT included adults aged 50 years or older who were at modest risk of vascular disease but who were free of diabetes mellitus and did not have a recent stroke, CV event, or heart failure. SPRINT showed that in this select population, lowering systolic BP to a goal of <120 mm Hg resulted in significantly lower rates of fatal and nonfatal CV events and death from any cause. The benefits of stricter BP goals were also confirmed in a prespecified subgroup analysis of 2636 subjects aged 75 years or older.¹⁷¹ Of interest, although the overall rate of serious adverse events was no different between groups (38.3% in intensive vs. 37.1% in the control group; $P = .25$), more patients in the intensive arm developed AKI, syncope, electrolyte abnormalities, and hypotension (but had no injurious falls). However, there is concern that the data from SPRINT may not be applicable to the older, frail patient population with advanced CKD.^{168,169,172} Although a proportion of those aged 75 years or older recruited in SPRINT had elevated creatinine levels (16% had an eGFR <45 mL/min/1.73 m²), few had other features of progressive kidney disease (see earlier discussion on controversies around defining older adults as having a disease based on eGFR thresholds alone).¹⁷¹ Patients with proteinuria >1 g/day and those with polycystic kidney disease were actively excluded. Unlike many CKD patients, BP was easily controlled with a minimal

number of medications. Furthermore, patients with dementia, unintentional weight loss >10%, or residence in a nursing home were excluded, as were individuals with medical conditions limiting life expectancy to <3 years (including cancer).

The benefits of treating hypertension have been confirmed in a meta-analysis incorporating data from almost 45,000 patients across all ages.¹⁷³ The evidence for which BP targets to use is less clear. Weiss and colleagues¹⁶⁸ have shown robust evidence that BP targets of less than 150/90 mm Hg led to a reduction in mortality, cardiac events, and stroke but only low to moderate evidence that a BP lower than 140/85 mm Hg was beneficial in adults aged 60 years or older. In addition, they found lower BP targets to be associated with an increase in syncope and medication burden but not higher rates of falls or cognitive decline. Although the rate of adverse effects seemed similar across all BP targets, the impact of an adverse event is likely higher in older patients, particularly those residing in a nursing home,¹⁷⁴ and consequently caution is advised. In 2021, the STEP study group published a trial targeting specifically older patients with hypertension. The group enrolled 8511 patients in either an intensive treatment (systolic BP target of 110 to <130 mm Hg) or a standard treatment (systolic BP target of 130 to <150 mm Hg).¹⁷⁵ The cardiovascular primary outcomes occurred less in the intensive group, but safety and renal outcomes did not differ. In all previously mentioned trials, comorbid patients including those with advanced CKD, frailty, and impaired cognition were excluded. Thus the reader is best advised to integrate data, as they emerge, with the results of a CGA into decision making and adapt treatments to align with individual patient complexity.

THE ROLE OF STATINS IN OCTOGENARIANS

CKD is associated with an increased risk of major CV events and mortality and is sometimes considered a coronary risk equivalent to diabetes and hypertension.¹⁷⁶ Although dyslipidemia is an important modifiable risk factor for CV disease in the general population, particularly in those older than 50 years, its role and relevance in the older CKD population are less clear. Current KDIGO guidelines recommend the initiation of statin therapy, either alone or in combination with ezetimibe, for all nondialysis patients aged 50 years or older with an eGFR <60 mL/min/1.73 m², irrespective of age or baseline lipid profile.¹⁷⁷ Much of the evidence is based on high-quality data from the Study of Heart and Renal Protection (SHARP).¹⁷⁸ The SHARP trial, published in 2011, randomized patients with CKD to 20 mg simvastatin plus 10 mg ezetimibe or placebo. Patients were followed prospectively for a median of 5 years, and results showed a lower incidence of cardiac death, stroke, nonfatal myocardial infarction, and cardiac revascularization (11.3% vs. 13.4%, respectively). Numerous participants were aged 70 years and older (28% of SHARP participants), and the effectiveness of the interventions, simvastatin and ezetimibe, was similar. There was little evidence from the data presented that suggested any increased risk of adverse events in these older participants. The dilemma, however, arises from what and who were not included in the study. Many patients were excluded because their physician did not think that participation was appropriate. Many of these may have been

older individuals with social or other characteristics not captured in the study data. In an eloquent piece, Butler and colleagues¹⁷⁹ have argued that application of current KDIGO guidelines advocating for the use of statins in those of any age with CKD may not be appropriate. They suggested that the older patients recruited into the study were likely not representative of many of those attending our clinics; they referred specifically to patients with a history of myocardial infarction, cognitive impairment, or functional limitations who would have been excluded but also allude to those with social or psychological frailty. Also, they remind us of the need to elicit treatment preferences and goals, especially because both pill burden and cost are often more important to older individuals than to younger patients, and the impacts of any adverse drug effects are higher.

Contemporary guidelines for the general population, published by the American College of Cardiology and American Heart Association,¹⁸⁰ include age-based criteria while the European Society of Cardiology guidelines recommended use of risk scores that incorporate competing risk.¹⁸¹ However, KDIGO kidney guidelines do not specify if age adaptations are warranted, and it may be appropriate to take a more cautious approach with both the use of statins and the peak dosage prescribed in older or frail persons in whom life expectancy is short. This issue remains debatable because some authorities still advocate for more aggressive lipid lowering.¹⁸²

While the benefits of lipid lowering have been shown for patients with nondialysis CKD, the benefits do not appear to extend to dialysis patients. Several large randomized trials and high-quality meta-analyses have suggested that statin therapy has little or no effect on cardiovascular outcomes among adults undergoing maintenance dialysis.^{178,183-189} Furthermore, emerging evidence has suggested that statins may accelerate vascular calcification in the dialysis population.¹⁹⁰ At present the 2014 KDIGO Lipid Guidelines¹⁷⁷ and 2021 Canadian Cardiovascular Society Guidelines¹⁹¹ recommend statins not be started in adults receiving dialysis but rather be continued in patients already receiving them at the time of dialysis initiation. Although not supported by evidence or by guidelines, discontinuation of statins may have a role in older individuals maintained on dialysis who present with increasing frailty, multimorbidity, polypharmacy, or malnutrition and those entering the end of life.

ANTICOAGULATION IN ATRIAL FIBRILLATION

The prevalence of atrial fibrillation (AF) in patients with impaired kidney function is considerably higher than that seen in the general population.^{192,193} Adults aged 65 to 74 years who are undergoing maintenance dialysis have a prevalence of AF of 13%. This increases to 23% among those 85 years or older.¹⁹⁴ A large proportion of these patients are treated with oral anticoagulants with the intention to reduce the risk of stroke and systemic embolism.

There is increasing evidence that the benefit-risk ratio of anticoagulation with warfarin worsens as kidney disease progresses, and some estimate bleeding risk to be 2 to 3 times higher in those with eGFR <30 mL/min/1.73 m² compared with those with normal kidney function. Ongoing physician uncertainty has led to warfarin no longer being the preferred anticoagulant of choice. Direct

oral anticoagulants (DOACs) are playing an increasing role, particularly in those with CKD stage 3/4a. According to a cohort of 2453 patients with nonvalvular atrial fibrillation and CKD 3 to 5 (dialysis patients were excluded), bleeding risk is reduced by a third in those taking DOACs compared with warfarin.¹⁹⁵ In a large cohort study of 115,564 patients with incident AF, 3% of 2940 patients with CKD had bleeding with DOAC use and after adjustment for additional comorbidity, bleeding risk approximated that of non-CKD individuals.¹⁹⁶ Current KDIGO guidelines suggest DOAC use in those with early CKD and lower-dose apixaban (2.5 mg orally twice daily) or rivaroxaban (15 mg daily) for those with eGFR <30 mL/min/1.73 m² until clinical safety data are available.¹⁹⁷ Apixaban remains the preferred drug for older adults due to its lower risk of major bleeding and gastrointestinal bleeding when compared with other DOACs. Studies comparing outcomes in the dialysis population, with and without anticoagulation, are awaited (www.clinicaltrials.gov NCT03987711 and NCT03862859), and therefore no recommendation can be made for those maintained on dialysis. Nevertheless, the use of these drugs has become more prevalent, with an estimated 12% of dialysis patients on anticoagulation being treated with DOAC.^{198,199}

ANTIRESORPTIVE TREATMENT: BONE HEALTH

Bone health remains a challenging aspect of kidney care, amplified in the older population by the attendant high risk of fragility fractures, adynamic bone disease, and polypharmacy. In this section, we review the data (or lack thereof) supporting the use of antiresorptive therapies in older patients who have recently had a fracture.

Fragility fractures, fractures that occur as a result of low-energy trauma such as a fall from standing height or lower, are common in older people and are associated with excess mortality, prolonged rehabilitation, long-term pain, and high medical costs. This is particularly the case with hip fractures, which have a substantial impact on an older individual's health, QOL, and abilities, with only 50% of patients regaining their prefracture level of mobility and independence.²⁰⁰ In 2019 there were an estimated 178 million new fractures worldwide (an increase of 33% since 1990).²⁰¹ The risk of fracture increases with CKD progression, resulting in risks much higher in older CKD individuals versus those with normal kidney function.²⁰² Hip fractures occurring in dialysis patients are particularly concerning because the 1-year mortality after hip fracture in the dialysis population is close to 65% versus 20% in the general U.S. population.²⁰³

Despite these alarming numbers, there are sparse data to guide clinicians providing postfracture care to older patients with CKD. Older patients with CKD stages 4 to 5D have been uniformly excluded from clinical trials evaluating pharmacotherapies. The uniqueness and complexity of CKD in the pathogenesis of fractures, mixed with age-related and postmenopausal causes of bone loss, render this population challenging in terms of diagnosis and management of osteoporosis. Active management of hyperparathyroidism may have an effect on fracture risk but, together with phosphate-lowering therapies, may also negatively affect other predisposing risks such as polypharmacy, malnutrition, and aging itself.

There have been few observational or retrospective studies regarding the use of bisphosphonates or denosumab in patients with CKD stages 4 to 5D. Clinical experience is limited, and the theoretic risks of administering antiresorptives must be carefully weighed against unknown benefits, particularly because bone pathology can be difficult to determine accurately. Widespread use of osteoporosis therapies in patients with advanced CKD and/or evidence of CKD-mineral and bone disorder (MBD) was previously discouraged, especially in frail older individuals.²⁰⁴ However, there are a few studies proving their harm or benefit and recent data suggest that bisphosphonates might be safe in patients with eGFR of 15–59 mL/min/1.73 m² due to age-related decline in kidney function without CKD-MBD.^{205,206}

Antiresorptives and other osteoporosis therapies can be safely used in patients with modest reductions in eGFR (≥ 30 mL/min/1.73 m²), particularly if the parathyroid hormone level lies within the normal to high range.²⁰⁷ However, for individuals with severe CKD (stages 4–5D) there are limited therapeutic interventions known to reduce fracture recurrence. Furthermore, the theoretic risks of adverse events remain of concern; for example, antiresorptives may worsen low bone turnover, osteomalacia, and mixed uremic osteodystrophy and exacerbate hyperparathyroidism, and denosumab may induce significant hypocalcemia.²⁰⁸ In the 2017 KDIGO guideline update, treatment recommendations are discretionary and recommended only after “taking into account the magnitude and reversibility of the biochemical abnormalities and the progression of CKD, with consideration of a bone biopsy.”²⁰⁷ Nonpharmacologic intervention with proven efficacy should always be part of the overall management of osteoporosis and includes vitamin D₃ supplementation, a reduction in alcohol intake, smoking cessation, fall prevention, improved nutrition, and weight-bearing exercise.

Nephrologists should continue to pay careful attention to emerging information in this relatively new field while prospective studies, specifically designed for CKD patients, are performed.

VASCULAR ACCESS CREATION

Vascular access creation in older patients undergoing HD remains a topic of ongoing discussion, particularly because of the need to balance concerns about infection-related morbidity and mortality against any potential residual confounding arising from the observational nature of most studies.^{209–212} There is a clear association with poor functional status and a lower use of AVF, suggesting earlier studies comparing AVFs with arteriovenous grafts (AVGs) or central venous catheters (CVCs) may have overestimated the survival benefits for those dialyzed via an AVF. By comparing outcomes of patients who underwent an unsuccessful fistula attempt with those who did not have fistula creation attempted, it appears that the observed survival gains are attributable to residual confounding factors, arising from the inherent bias that individuals undergoing surgery are likely healthier than those for whom surgery is not attempted.^{213,214} Furthermore, although the relative gains may be similar regardless of age, the absolute gains, in terms of survival days, are likely to be small because overall survival on dialysis is lower in older individuals. One argument in favor of avoiding CVC use is that *Staphylococcus aureus* bloodstream

infections are known to be 10-fold higher in individuals undergoing HD with central vein catheter use (adjusted rate ratio 6.2; 95% CI = 5.7–6.7 for dialysis catheter vs. fistula).²¹⁵ This has been challenged by DeSilva and colleagues,²¹⁶ who analyzed data from 115,425 incident U.S. patients on HD aged 67 years and older. They assessed mortality outcomes on the basis of the first vascular access placed but found no difference in mortality for patients with an AVG as the first access compared with those who started with an AVF. Similar results were reported by Drew and colleagues.²¹⁷ They used decision analysis to apply current knowledge of the risks and benefits to different access strategies. Although their study has faced criticism for assuming a base where patients initiate dialysis with a CVC, the methodology used is robust, valid, and generalizable. Results show a modest survival benefit for those aged 80 years when patients are dialyzed using a fistula strategy. Maximal life expectancy gains were seen in males and nondiabetic patients. For example, an 80-year-old nondiabetic male had an estimated life prolongation of 8 months with a fistula rather than CVC and 6 months when compared with AVG. Survival, however, for an 80-year-old diabetic female was best with an AVG and, when comparing AVF with CVC, survival differed by only 3 weeks.²¹⁷

Access-related morbidity is also known to be high. Observational studies suggest that up to one-third to one-half of octogenarians undergoing predialysis fistula creation die before they need kidney replacement therapy (RRT).^{216,218} Fistula failure rates are high in older patients, with some studies suggesting up to 60% of older adults will have a fistula that fails to mature.^{219–221} Of those that do function, several will require multiple interventions, particularly early in the clinical course, causing a high clinical and economic burden when overall survival and gains are limited.²¹² Patients undergoing dialysis with an AVF or an AVG are more likely to report pain, bleeding, bruising, and swelling than those with a CVC.²²² Cannulation-related complications are more common, in part due to fragile skin and in part related to the higher use of antiplatelet or anticoagulant medications.²²³ Access creation is also associated with reduced grip strength and impaired arm and hand function.²²⁴ In patients with marginal functional status, creation of AVFs or AVGs may affect personal independence and limit dressing, feeding, and other activities of daily living, although bathing and showering were considered more difficult with CVC.²²⁵ Most clinicians would agree it is ultimately important to align the choice of dialysis access to individual patient expectations.²²⁶

The ideal clinical scenario is when results of the individual's CGA help inform the choice of vascular access.²²⁷ AVGs in predialysis patients placed closer to the time of HD initiation have been shown to be an effective CVC-sparing strategy²²⁸ and to be less costly than placing an AVF.²¹² Because of this, we continue to recommend a patient-first approach that identifies “the right access for the right patient” based on patient preferences, priorities, life expectancy, and QOL.

CARE FOR OLDER ADULTS WITH END-STAGE KIDNEY DISEASE

DIALYSIS

The demographics of the dialysis population continue to change with time. Compared with the early 2000s, patients

are older and more medically complex²²⁹ with 42% being older than 65 years and 17% older than 75 years.²³⁰ Until recently, the adjusted ESKF prevalence has also been rising, with steeper increases among older age groups compared with those younger than 65 years. These trends, however, have started to change, and data from the U.S. Renal Data System have suggested more conservative dialysis practice patterns, with pronounced declines in the number of older people starting dialysis resulting in 2020 incidence rates being the lowest among those aged 65 years and older since the late 1990s.²³⁰

Although it is often assumed that treatment with dialysis will extend life and alleviate the signs and symptoms of advanced kidney disease, there is growing evidence that benefits may not always accrue in older adults. Longitudinal studies, across much of Europe, have shown that the symptom burden increases dramatically in the 6 months preceding dialysis initiation, with a corresponding decline in QOL

(Fig. 60.5).^{231,232} The recovery for QOL scores, after dialysis initiation, is, however, modest. Likewise, few individuals report a return to predialysis symptom burden.²³³

Life extension is also modest. The median life expectancy after the initiation of chronic dialysis in the United States is comparable with (or worse than) many cancers and estimated to be less than 2 years for patients aged 75 years or older, although considerable heterogeneity exists. The data suggest that up to 25% of those aged 70 years or older die within the first 6 months. Conversely, a similar proportion survive beyond 4 years.^{76,96} For all ages, the relative median life expectancy after starting dialysis is around 25% to 30% that of age-matched individuals from the general population,²³⁴ and therefore older adults are most likely to have the smallest absolute gain measured in months or years.

Among factors that predict early mortality are the presence of frailty, impaired functional status, low body weight or

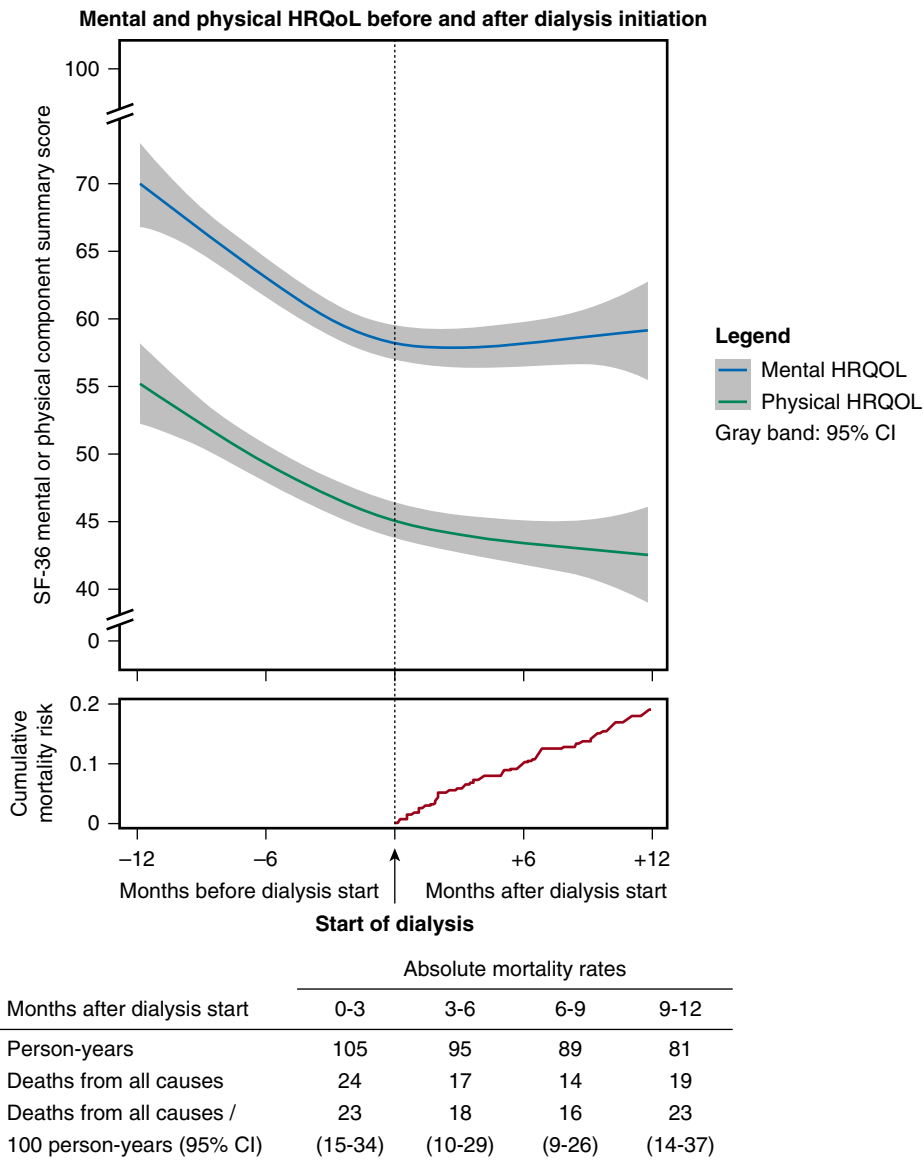


Fig. 60.5 Figure showing change in quality of life score in older adults before, and after starting dialysis. (From de Rooij ENM, Meuleman Y, de Fijter JW, et al. Quality of life before and after the start of dialysis in older patients. *Clin J Am Soc Nephrol.* 2022;17(8):1159–1167.)

serum albumin concentration, and need for nursing care at the time of dialysis initiation.^{90,95,235-239} There are several prognostic tools,^{92,93,235,240,241} but most are limited to use at the time individuals initiate dialysis and do not help prognosticate survival in those who are already established on dialysis or who opt for a comprehensive conservative kidney care approach. The limits and deficiencies of the most common prognostic tools have been reported elsewhere.²⁴²

DIALYSIS MODALITY SELECTION

Among patients 65 years of age and older initiating dialysis in the United States, most of them (87%) will start in-center HD as their initial modality choice, followed by PD (11%) and preemptive transplantation (1.6%).²³⁰ Until recently, one of the main arguments favoring HD was based on the overall mortality rates seen in the older patient population treated with PD. In a meta-analysis involving 631,421 older patients, the mortality was higher in those undergoing PD compared with HD (HR, 1.10; 95% CI, 1.01–1.20).²⁴³ However, particularly within the field of geriatric nephrology, it is important to recognize that the flexibility and gentle nature of PD offers much to an older individual, and many individuals forfeit a longer time alive but feeling unwell for a shorter time feeling well. Patients may be attracted to several other aspects of PD, such as increased dietary freedom, decreased hospital visits, control over one's own care, and a feeling of personal autonomy.²⁴⁴ In those with a guarded prognosis but high symptom levels from fluid accumulation, PD may be an ideal alternative to comprehensive conservative kidney care because it can be adapted to patients' needs as they transition along their disease course until they reach end of life.²⁴⁵ For example, in a patient experiencing increasing pain or deterioration

from non-kidney-related comorbidity, dialysis exchange frequency can be reduced and the solution volume and concentration adapted to improve comfort, prevent shortness of breath, and enhance the feeling of control and freedom.

Frail older patients who are unable to perform their own PD exchanges can be successfully supported by assisted PD.^{246,247} With assisted PD, home care workers, family members, or staff in long-term care facilities are trained to perform the exchanges and supervise the dialysis. The level and tasks for assistance vary and assistance tends to be required long term (Fig. 60.6).^{248,249} Different models have been used across different countries, and currently neither functional dependence nor cognitive impairment are considered absolute barriers to PD.^{248,250-252}

In-center HD may be preferred over home-based PD treatment by some individuals because their day can be structured around scheduled and fixed times, facilitating coordination for family members or staff providing support. Others benefit from socialization with staff and other patients at the dialysis facility and from having frequent contact with the dialysis staff to provide both medical and psychosocial support.^{244,253} HD in frail older patients is, however, more likely to precipitate episodic postural hypotension and possibly accelerate cognitive decline from myocardial and cerebral stunning.²⁵⁴ Additional challenges include the creation and maintenance of vascular access; problems associated with postdialysis recovery time, such as fatigue; and the risk of falls after dialysis.²⁵⁵

As a result, there is no one dialysis modality of choice for the frail older patient. Clinicians are advised to use information from the CGA to build a framework of total care, focusing on the needs of the individual patient and the goals of care in the context of her or his life experience.

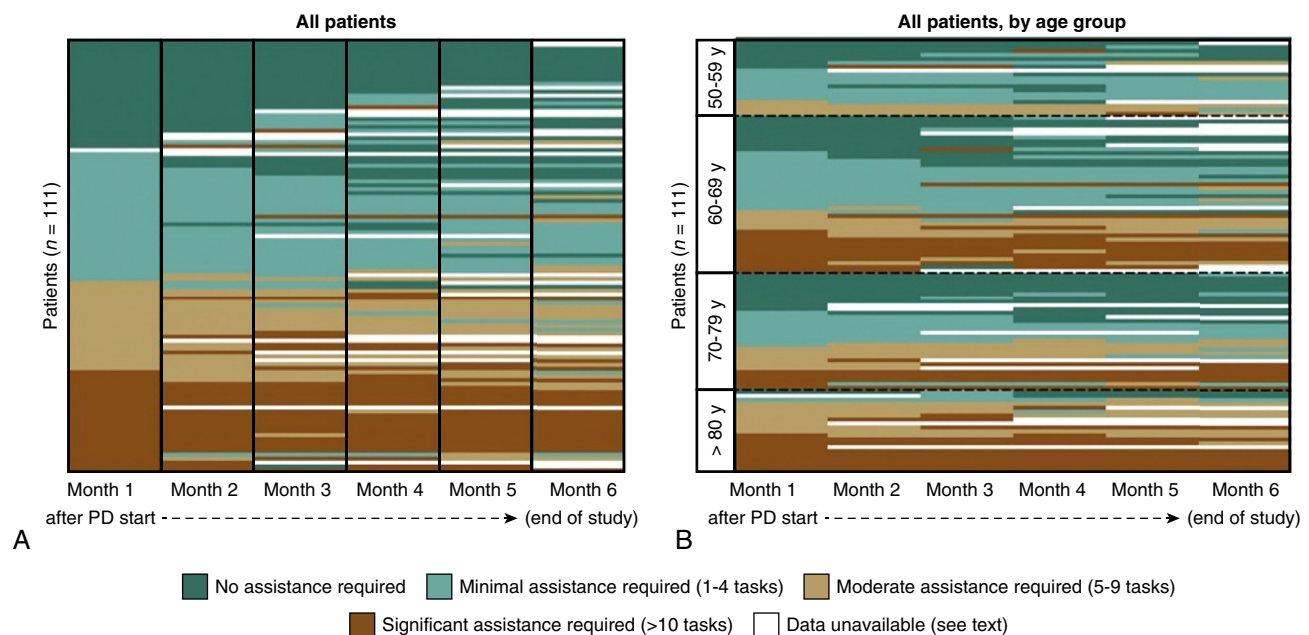


Fig. 60.6 (A and B) The change in need for assistance over time in older patients maintained on dialysis. The different shading patterns represent the degree of assistance provided. (From Fonseca-Correa JI, Farragher JF, Tomlinson G, et al. Longitudinal changes in the use of PD assistance for patients maintained on peritoneal dialysis. *Kidney360*. 2021;2(3):469–476.)

KIDNEY TRANSPLANTATION

In the United States, patients aged 65 years or older currently comprise 26% of all kidney transplant wait-listed patients; 21% of all recipients are older adults. While in general the number of transplants occurring per year has remained relatively static, there has been a progressive increase in the number of individuals aged 65 years or more undergoing transplant surgery, with the absolute number of older patients increasing almost twofold between 2008 and 2021.²⁵⁶ Although advanced age is no longer a barrier to kidney transplantation, age remains an important surrogate factor when assessing suitability for transplantation. Chronologic age is less important than the patient's level of frailty, burden of coexisting conditions, and overall prognosis.^{257,258} Frailty assessment is increasingly recognized as an essential component of pretransplant assessment to identify and exclude high-risk candidates and allow transplantation to elderly patients regardless of age.^{259,260} However, the best modality for frailty assessment in this context has not been yet defined.²⁶¹ Transplantation assessments are also commonly more rigorous for older candidates when evaluating cardiac and cancer risk.²⁶²

Transplantation offers survival benefits across all ages, although due to the attendant risks of the surgery itself and side effects associated with immunosuppressive drugs, patients have an increased mortality risk in the initial few months. Survival equivalence, when compared with maintenance dialysis, is seen approximately 8 months after the surgery is performed in younger patients but not seen in those aged 60 years and older until more than 12 months have elapsed from the date of transplantation.²⁶³ In those aged 70 years and older at the time of transplantation, survival advantage may be delayed beyond 21 months post transplantation.²⁶⁴ Several strategies have been used to maximize kidney transplantation accessibility including the use of expanded criteria donor (ECD) kidneys and pairing organs harvested from older donors with older recipients.²⁶⁵ These systems offer advantages and disadvantages, but overall the most beneficial and commonly used strategy in North America is to offer ECD kidneys to the older patient.²⁶⁶

The most common cause of graft loss in older kidney transplant recipients remains death with a functioning graft.^{267,268} Death-censored graft failure risk remains lowest, across all age groups, in patients aged 65 years or older likely because of a shorter life span and lower risk of acute rejection. Older patients are at higher risk of medication-related adverse events and higher rates of opportunistic infections and neoplasia.^{269,270}

COMPREHENSIVE CONSERVATIVE KIDNEY CARE

Although the change in dialysis utilization is captured across several regions of the world through dialysis registries, little is known about how often patients with advanced stages of CKD choose not to undergo dialysis or are not offered

dialysis. Single-center studies from Europe and Australia have suggested that a substantial number of older patients with advanced kidney disease are treated conservatively and do not receive dialysis.²⁷¹⁻²⁷⁶ Other studies, using administrative data, also found that a notable number of older patients do not undergo dialysis. These studies identified patients with advanced CKD who were likely at risk of dialysis by selecting those who had two or more creatinine levels determined at least 3 months apart that met criteria for a low eGFR. In one study, from Canada, the clinical course of all individuals with creatinine values consistent with CKD were tracked over a 2-year period.²⁷⁷ In the second study, from the Veterans Administration, detailed data were extracted from electronic chart data and used to identify whether patients either underwent dialysis or were being prepared for dialysis.²⁷⁸ In both studies, older adults were found to be less likely to undergo dialysis than their younger counterparts, implying that there may be a relatively large “reservoir” of older adults with high creatinine levels (and, by implication, low levels of eGFR) who, for a number of reasons, do not get started on dialysis (Fig. 60.7). Similar patterns have been reported for the incidence of hospitalized patients with AKI treated with dialysis.²⁷⁹ Although provocative, these data cannot distinguish between those who did not meet clinical criteria or a need for dialysis from those who did not get offered or who chose not to undergo dialysis.

Although registry data serve as a valuable source of information about outcomes in older adults with ESKF who are treated with dialysis, less is known about outcomes among those who are not treated with dialysis. What little is known about this group comes from several small, single-center studies that have examined outcomes among patients with advanced kidney disease who were managed with comprehensive conservative kidney care (CCRC) (see also Chapter 61).

Most studies have shown that dialysis is associated with a modest survival advantage over CCRC; however, the magnitude of this effect varies across studies, and the potential advantage seems to be lost for a subgroup of older patients with a high burden of comorbidity.²⁸⁰⁻²⁸² The comparisons, however, between patients managed with CCRC and those initiating dialysis are challenging because there is no clear starting point for analysis. It is often difficult to know when dialysis would have been started in those who chose a CCRC pathway, and it is possible that studies suffer from a lead time bias. Robust patients and those with rapid declines in kidney function are more likely to undergo dialysis; those who are frail or have slowly progressing kidney declines are more likely to undergo CCRC. Thus the use of time points determined from the eGFR, to set a time when dialysis may have been otherwise started, is likely to lead to further systematic biases.

Evolving literature has recognized that many older patients may benefit equally from dialysis as from CCRC. Particularly in older adults with a high burden of GSs, small gains in longevity that are potentially achieved with dialysis care need to be balanced against the high risk of increased symptom burden, worsening dependency, and often a

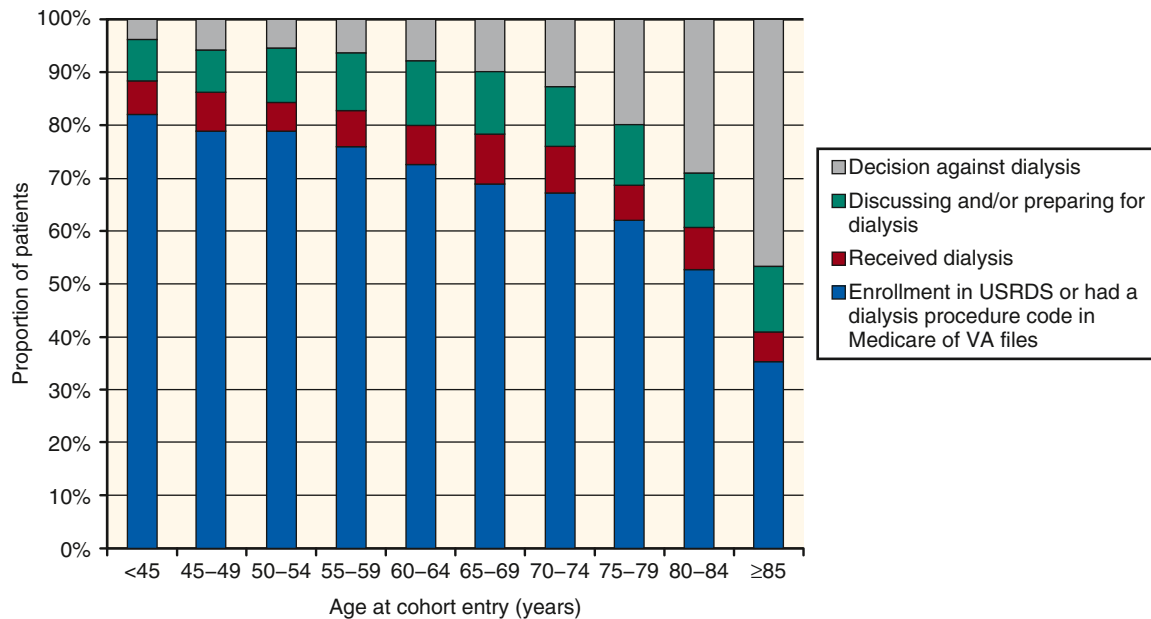


Fig. 60.7 Age differences in treatment decisions and practices for advanced kidney disease. Based on a chart review of a random sample of patients not enrolled in the U.S. Renal Data System and who did not have a dialysis procedure code in Medicare or Department of Veteran Affairs (VA) administrative files during follow-up. (From Wong SP, Hebert PL, Laundry RJ, et al. Decisions about kidney replacement therapy in patients with advanced kidney disease in the US Department of Veterans Affairs, 2000-2011. *Clin J Am Soc Nephrol*. 2016;11(10):1825-1833.)

negative effect on QOL.^{255,283,284} Overall gains are modest, particularly in those who have recently undergone one or more life-prolonging procedures, such as intubation, feeding tube placement, and cardiopulmonary resuscitation.²⁸⁵

Patients who opt for a CCRC pathway tend to fare well for several months. Their symptom burden and QOL remain stable until the last 1 or 2 months of life, at which time palliative care interventions are often beneficial.²⁸² Functional status is often maintained until the last months of life and, overall, patients are more likely to die in their home environment or in hospice compared with those starting RRT.²⁸⁰ Further information on the trajectories seen in those initiating dialysis have been emerging, although insufficient information is currently available to make clear distinctions between patient groups.

IMPORTANCE OF SHARED DECISION MAKING IN END-STAGE KIDNEY DISEASE

Shared decision making is a critical component of kidney care, particularly for older patients and their families. Shared decision making incorporates discussions about realistic treatment options with patients and their families or caregivers in open, collaborative, and structured environments (see [Chapter 61](#)). Discussions should be specific to the individual and his or her lifestyle and values, as well as limited to medically appropriate interventions. Ideally, a multidisciplinary kidney team, patient, and family should be present and information should be shared about the potential benefits and harms of various treatment options. Prognosis should be integrated with the results of the CGA, allowing clinicians to share information

about possible positive and negative effects on daily living and symptom burden. Information about longevity, the expected changes in symptom burden, and improvement in social functioning can be customized to the individual ([Fig. 60.8](#)).²³³ This approach, in which patients and families are involved in an individualized management plan, will maximize treatment benefits.

DIALYSIS TREATMENTS UNIQUE TO OLDER PATIENTS

DIALYSIS AND GERIATRIC REHABILITATION PROGRAMS

Geriatric rehabilitation programs are designed to optimize the functioning of older adults and often pre-morbidly frail individuals who have experienced loss of independence due to acute illness or injury, often superimposed on chronic functional and medical problems.²⁸⁶ Geriatric rehabilitation provides evaluative, diagnostic, and therapeutic interventions to restore functional ability or enhance residual functional capacity in older people with disabling impairments.²⁸⁷ Physical deterioration occurs in predialysis CKD before commencement of RRT.^{288,289} Once patients start RRT, they appear to decrease their levels of physical activity further,²⁹⁰ leading to increased skeletal muscle wasting and poorer overall physical function.^{291,292} This vulnerable state is exacerbated by multiple or recurrent periods of illness or hospitalization, which, even if short-lived, may lead to persistent disability.²⁹³ Over a span of 6 months, vulnerable individuals experience increased dependency in a number of aspects of daily living, which may eventually result in a nursing home placement.^{76,77,294}

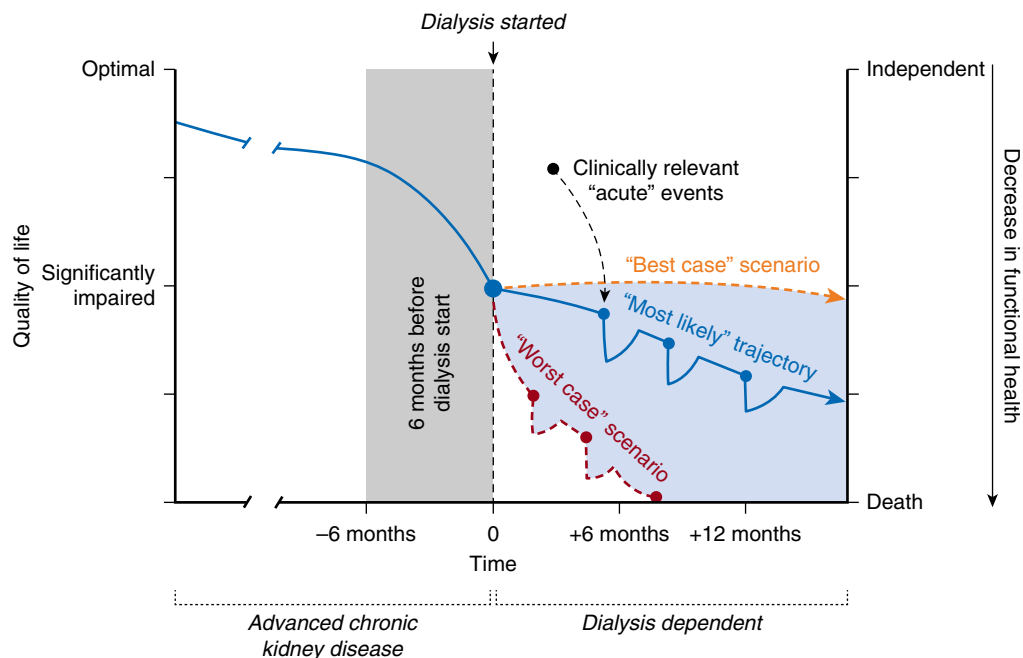


Fig. 60.8 Hypothetic life trajectory in older patients who start dialysis treatments across a spectrum of outcomes. (From Fonseca-Correa JI, Jassal SV. Health care for older adults with kidney failure. *Clin J Am Soc Nephrol.* 2022;17(8):1110–1112.)

Timely referral for geriatric rehabilitation may reverse recent-onset dependence and disability and increase the likelihood that patients return home or to a community residential setting after rehabilitation.^{295,296}

Specialized dialysis rehabilitation programs integrate active rehabilitation with dialysis care. An interdisciplinary team consisting of multiple professionals including physicians, nurses, pharmacists, physiotherapists, occupational therapists, speech-language pathologists, social workers, psychologists, dietitians, and recreation therapists work together, often with overlapping roles, to provide interventions such as therapeutic exercises, fall prevention plans, education on ways to conserve energy during activity, and recommendations of assistive devices. Geriatric dialysis rehabilitation often occurs at a turning point in a patient's disease trajectory and can allow a patient to recover personal function after a complicated course in a hospital or a recent multisystem illness. It is also an opportune moment to initiate or pursue discussions around goals of care and advance care planning. Providing dialysis in short daily sessions, if feasible, may improve the success of the program. Short daily dialysis (6 times/week for 2 hours) is well tolerated, limits scheduling conflicts and interference between medical and rehabilitation treatments, and may lead to improved nutrition and better participation in therapy sessions. Patients report less fatigue and fewer symptoms associated with rapid fluid shifts.²⁹⁷

Limited effectiveness data exist, with the largest information coming from a single-center report of 164 patients. In this series, more than 70% of patients met functional goals and showed clinically meaningful improvements in personal independence.^{295,298} Other models of geriatric rehabilitation including outpatient

rehabilitation, community rehabilitation, and fall prevention programs, have not been well studied in the older kidney population but have been shown to benefit other clinical geriatric populations and thus warrant further investigation in the CKD population. A practical first step is for nephrology teams to consider if rehabilitation would be beneficial using information gathered through the CGA—for example, at dialysis initiation, after hospitalization or accidental fall(s), or if there is a change in social status (e.g., changing residence or the death of a caregiver or key family member). Nephrologists may collaborate with the rehabilitation team to reevaluate health targets at this time.

PALLIATIVE DIALYSIS

There has been an emerging dialogue around the role of a palliative approach to care for those on maintenance dialysis.^{276,299-302} The palliative approach to dialysis care can be defined as a transition from a conventional disease-oriented focus on dialysis as rehabilitative treatment to an approach prioritizing comfort and alignment. In the latter approach, patient preferences and goals of care are considered to improve QOL and reduce symptom burden for maintenance dialysis patients in their final years of life.³⁰⁰

One approach is to classify patients into one of three treatment groups, each with a different expected outcome:

1. Dialysis as a bridging treatment or long-term maintenance treatment, when the patient is expected to return to usual life activities
2. Dialysis as a final destination, when dialysis treatment is used to alleviate symptoms but CKD and/or non-CKD disease are expected to continue to deteriorate
3. Active medical management without dialysis³⁰³

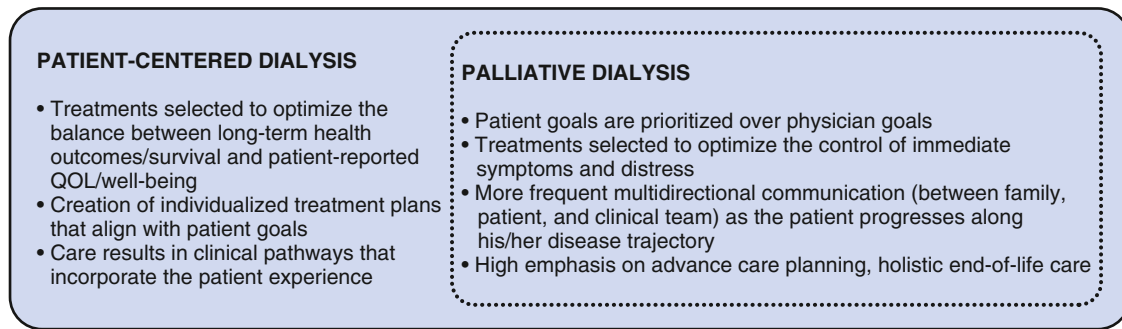


Fig. 60.9 Conceptualizing palliative dialysis. In patient-centered dialysis, treatments are chosen to align with patient goals. In palliative dialysis, a subset within patient-centered care, treatments are chosen so that if needed, patient goals are prioritized over physician goals. QOL, Quality of life.

Those with high levels of comorbidity, ongoing concomitant nonkidney disease, or an unclear prognosis likely best fit within the second final destination group and are those who are most likely to benefit from a palliative approach to dialysis care. After shared decision making, dialysis treatments can be customized to alleviate the individual's symptoms and minimize treatment-related distress.³⁰⁴

Palliative dialysis is often, incorrectly, thought of as being equivalent to less dialysis or a precursor to dialysis discontinuation. Although it is often necessary to adapt dialysis timings and prescriptions, and dialysis discontinuation may be a component, these interventions are rarely sufficient to ameliorate symptoms and suffering.²⁹⁹ Ongoing symptom assessment and therapeutic interventions (discussed briefly later) are required. When taking a palliative approach to kidney care, advance care planning becomes of paramount importance. As the patient continues along the disease trajectory, he or she, together with the immediate family, will increasingly need support as they learn to accept death as a natural consequence. As death approaches, the focus becomes more consistent with interventions that improve the dying process and end-of-life care. At this point, patients' goals of care tend to shift to focus almost exclusively on QOL and minimizing suffering rather than survival, with a strong emphasis on emotional, social, and family support.³⁰⁵ Palliative dialysis is a form of patient-centered dialysis in that it continues to align care with patient preferences. In palliative dialysis, the focus of care is on the prevention and relief of suffering. Treatments are adapted to the issues that matter most to the patient and are adjusted so that the individual patient can achieve goals for the remaining life. This approach, also termed "patient-centered care," requires setting a balance between issues important to the clinician (e.g., survival and laboratory targets) with those affecting the individual (restrictions affecting QOL, symptom control, and physical function; Fig. 60.9).²⁹⁹ The clinician may therefore need to forgo traditional treatment targets and disease-oriented guidelines in favor of the patient's symptom management and individualized goals of care. This can mean that less emphasis is placed on maximizing long-term health outcomes, such as optimal control of BP, phosphate levels,

and other laboratory results, unless they result in immediate symptom improvement.

Common symptoms associated with advanced CKD include fatigue, nausea and vomiting, sleep disorders, restless legs syndrome, depressive symptoms, and pruritus.²⁹⁹ Many occur in complex clusters, and treatments are primarily to ease symptoms sufficiently to maximize comfort because it is not always possible to alleviate them completely.^{283,284} Less dialysis rarely provides benefits and may result in increased symptoms and postdialysis fatigue, especially if greater ultrafiltration is required to manage troublesome shortness of breath. However, it can be an option for those in whom fluid management is not a big problem and who favor spending more time at home, with reduced transportation to and from the dialysis facility over symptoms related to underdialysis. Engaging patients and families in discussions often leads to identification of the issues and symptoms that lead to a request to reduce dialysis. Some patients will feel stressed when rushing to prepare for an early morning dialysis session or when they must change their routine to eat their midday meal earlier and their evening meal later when attending afternoon dialysis. Dosing and timing of dialysis can then be adjusted accordingly to improve dialysis tolerance. Numerous symptoms can also be mitigated by more frequent, but shorter, dialysis sessions, such as sleep disorders, respiratory distress, and postdialysis fatigue. This option may be acceptable for some patients who are not distressed by the dialysis procedure itself.

CONCLUSION

Increasingly, the care of older patients with CKD is being recognized as a subspecialty within nephrology. Clinicians must balance efficient, high-quality medical care with a personalized and humane approach to the older individual. As our knowledge within geriatric nephrology grows, we are entering a new era with personalized, quality care. At the core of this care is the need to have a better and deeper understanding of the individual to whom we are providing care so that any treatments or interventions we suggest align best with these needs.

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KEY REFERENCES

1. Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152(9):561–567.
7. Glasscock R, Delanaye P, El Nahas M. An age-calibrated classification of chronic kidney disease. *JAMA.* 2015;314(6):559–560.
10. Beridze G, Vetrano DL, Marengoni A, Dai L, Carrero JJ, Calderon-Larranaga A. Concordance and discrepancies among 5 creatinine-based equations for assessing estimated glomerular filtration rate in older adults. *JAMA Netw Open.* 2023;6(3):e234211.
11. Carnevale V, Tinti MG. Use of eGFR in older adults with kidney disease. *N Engl J Med.* 2022;387(6):573–574.
19. Schaeffner ES, Ebert N, Kuhlmann MK, et al. Age and the course of GFR in persons aged 70 and above. *Clin J Am Soc Nephrol.* 2022.
20. Delanaye P, Pottel H, Melsom T. Physiology of the aging kidney: we know where we are going, but we don't know how. *Clin J Am Soc Nephrol.* 2022.
21. O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol.* 2007;18(10):2758–2765.
39. Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA.* 2011;305(15):1553–1559.
45. Hallan SI, Matsushita K, Sang Y, Mahmoodi BK, Black C, Ishani A, et al. Age and association of kidney measures with mortality and end-stage renal disease. *JAMA.* 2.
52. Shlipak MG, Matsushita K, Arnlov J, et al. Cystatin C versus creatinine in determining risk based on kidney function. *N Engl J Med.* 2013;369(10):932–943.
56. Stuck AE, Siu AL, Wieland GD, Adams J, Rubenstein LZ. Comprehensive geriatric assessment: a meta-analysis of controlled trials. *Lancet.* 1993;342(8878):1032–1036.
58. Hall RK, Haines C, Gorbattin SM, et al. Incorporating geriatric assessment into a nephrology clinic: preliminary data from two models of care. *J Am Geriatr Soc.* 2016;64(10):2154–2158.
67. Bowling CB, Booth 3rd JN, Safford MM, et al. Nondisease-specific problems and all-cause mortality in the REasons for geographic and racial differences in stroke study. *J Am Geriatr Soc.* 2013;61(5):739–746.
68. Bowling CB, Muntner P, Sawyer P, et al. Community mobility among older adults with reduced kidney function: a study of life-space. *Am J Kidney Dis.* 2014;63(3):429–436.
76. Johansen KL, Delgado C, Bao Y, Kurella Tamura M. Frailty and dialysis initiation. *Semin Dial.* 2013;26(6):690–696.
85. Kurella Tamura M, Covinsky KE, Chertow GM, Yaffe K, Landefeld CS, McCulloch CE. Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med.* 2009;361(16):1539–1547.
86. Jassal SV, Chiu E, Hladunewich M. Loss of independence in patients starting dialysis at 80 years of age or older. *N Engl J Med.* 2009;361(16):1612–1613.
89. Jassal SV, Karaboyas A, Comment LA, et al. Functional dependence and mortality in the international dialysis outcomes and practice patterns study (DOPPS). *Am J kidney Dis.* 2016;67(2):283–292.
91. Tennankore K, Zhao J, Karaboyas A, et al. The association of functional status with mortality and dialysis modality change: results from the peritoneal dialysis outcomes and practice patterns study (PDOPPS). *Perit Dial Int.* 2019;39(2):103–292.
96. Gill TM, Kurland B. The burden and patterns of disability in activities of daily living among community-living older persons. *J Gerontol A Biol Sci Med Sci.* 2003;58(1):70–75.
108. Tinetti ME, Williams CS. Falls, injuries due to falls, and the risk of admission to a nursing home. *N Engl J Med.* 1997;337(18):1279–1284.
117. Plantinga LC, Patzer RE, Franch HA, Bowling CB. Serious fall injuries before and after initiation of hemodialysis among older esrd patients in the United States: a retrospective cohort study. *Am J Kidney Dis.* 2017;70(1):76–83.
128. Hedayati SS, Minhajuddin AT, Afshar M, Toto RD, Trivedi MH, Rush AJ. Association between major depressive episodes in patients with chronic kidney disease and initiation of dialysis, hospitalization, or death. *JAMA.* 2010;303(19):1946–1953.
137. Lopes AA, Albert JM, Young EW, et al. Screening for depression in hemodialysis patients: associations with diagnosis, treatment, and outcomes in the DOPPS. *Kidney Int.* 2004;66(5):2047–2053.
141. Nagler EV, Webster AC, Vanholder R, Zoccali C. Antidepressants for depression in stage 3–5 chronic kidney disease: a systematic review of pharmacokinetics, efficacy and safety with recommendations by European Renal Best Practice (ERBP). *Nephrol Dial Transplant.* 2012;27(10):3736–3745.
144. Hedayati SS, Yalamanchili V, Finkelstein FO. A practical approach to the treatment of depression in patients with chronic kidney disease and end-stage renal disease. *Kidney Int.* 2012;81(3):247–255.
148. Kantor ED, Rehm CD, Haas JS, Chan AT, Giovannucci EL. Trends in prescription drug use among adults in the United States from 1999–2012. *JAMA.* 2015;314(17):1818–1831.
157. Novick TK, Surapaneni A, Shin JI, et al. Associations of opioid prescriptions with death and hospitalization across the spectrum of estimated GFR. *Clin J Am Soc Nephrol.* 2019;14(11):1581–1589.
160. American Geriatrics Society Beers Criteria Update Expert P. American Geriatrics Society 2023 updated AGS Beers Criteria(R) for potentially inappropriate medication use in older adults. *J Am Geriatr Soc.* 2023.
161. Sekiguchi T, Kabayama M, Ryuno H, et al. Association between protein intake and changes in renal function among Japanese community-dwelling older people: the SONIC study. *Geriatr Gerontol Int.* 2022;22(4):286–291..
178. Weiss J, Freeman M, Low A, et al. Benefits and harms of intensive blood pressure treatment in adults aged 60 Years or older: a systematic review and meta-analysis. *Ann Intern Med.* 2017;166(6):419–429.
184. Williamson JD, Supiano MA, Applegate WB, et al. Intensive vs standard blood pressure control and cardiovascular disease outcomes in adults aged ≥ 75 Years: a randomized clinical trial. *JAMA.* 2016;315(24):2673–2682.
190. Baigent C, Landray MJ, Reith C, et al. The effects of lowering LDL cholesterol with simvastatin plus ezetimibe in patients with chronic kidney disease (Study of Heart and Renal Protection): a randomised placebo-controlled trial. *Lancet.* 2011;377(9784):2181–2192.
192. Hawley CE, Roefaro J, Forman DE, Orkaby AR. Statins for primary prevention in those aged 70 Years and older: a critical review of recent cholesterol guidelines. *Drugs Aging.* 2019;36(8):687–699.
213. Welander F, Renlund H, Dimeny E, Holmberg H, Sjalander A. Direct oral anticoagulants versus warfarin in patients with non-valvular atrial fibrillation and CKD G3–G5D. *Clin Kidney J.* 2023;16(5):835–844.
214. Bansal N, Zelnick LR, An J, et al. Association of kidney function with risk of adverse effects of therapies for atrial fibrillation. *Kidney Int Rep.* 2023;8(3):606–618.
225. Chan KE, Edelman ER, Wenger JB, Thadhani RI, Maddux FW. Dabigatran and rivaroxaban use in atrial fibrillation patients on hemodialysis. *Circulation.* 2015;131(11):972–979.
235. Wilson LM, Rebholz CM, Jirru E, et al. Benefits and harms of osteoporosis medications in patients with chronic kidney disease: a systematic review and meta-analysis. *Ann Intern Med.* 2017;166(9):649–658.
237. Toussaint ND, Elder GJ, Kerr PG. Bisphosphonates in chronic kidney disease; balancing potential benefits and adverse effects on bone and soft tissue. *Clin J Am Soc Nephrol.* 2009;4(1):221–233.
243. Vieceilli AK, Lok CE. Hemodialysis vascular access in the elderly: getting it right. *Kidney Int.* 2019;95(1):38–49.
256. O'Hare AM, Allon M, Kaufman JS. Whether and when to refer patients for predialysis AV fistula creation: complex decision making in the face of uncertainty. *Semin Dial.* 2010;23(5):452–455.
261. de Rooij ENM, Meuleman Y, de Fijter JW, et al. Quality of life before and after the start of dialysis in older patients. *Clin J Am Soc Nephrol.* 2022;17(8):1159–1167.
262. de Rooij ENM, Meuleman Y, de Fijter JW, et al. Symptom burden before and after dialysis initiation in older patients. *Clin J Am Soc Nephrol.* 2022;17(12):1719–1729.
263. Fonseca-Correa JI, Jassal SV. Health care for older adults with kidney failure. *Clin J Am Soc Nephrol.* 2022;17(8):1110–1112.
270. Couchoud C, Hemmelgarn B, Kotanko P, Germain MJ, Moranne O, Davison SN. Supportive care: time to change our prognostic tools and their use in CKD. *Clin J Am Soc Nephrol.* 2016;11(10):1892–1901.

274. Oliver MJ, Quinn RR. Selecting peritoneal dialysis in the older dialysis population. *Perit Dial Int*. 2015;35(6):618–621.
275. Farragher JF, Oliver MJ, Jain AK, Flanagan S, Koyle K, Jassal SV. PD assistance and relationship to Co-existing geriatric syndromes in incident peritoneal dialysis therapy patients. *Perit Dial Int*. 2019;39(4):375–381.
286. Segall L, Nistor I, Pascual J, et al. Criteria for and appropriateness of renal transplantation in elderly patients with end-stage renal disease: a literature review and position statement on behalf of the European renal association-European dialysis and transplant association descartes working group and European renal best practice. *Transplantation*. 2016;100(10):e55–e65.
288. Rao PS, Merion RM, Ashby VB, Port FK, Wolfe RA, Kayler LK. Renal transplantation in elderly patients older than 70 years of age: results from the Scientific Registry of Transplant Recipients. *Transplantation*. 2007;83(8):1069–1074.
304. Carson RC, Juszczak M, Davenport A, Burns A. Is maximum conservative management an equivalent treatment option to dialysis for elderly patients with significant comorbid disease? *Clin J Am Soc Nephrol*. 2009;4(10):1611–1619.
309. Wong SP, Kreuter W, O'Hare AM. Treatment intensity at the end of life in older adults receiving long-term dialysis. *Arch Intern Med*. 2012;172(8):661–663; discussion 3–4.
320. Farragher J, Einbinder Y, Oliver MJ, Nachman J, Chow E, Jassal SV. Importance of early inpatient geriatric rehabilitation on outcomes in individuals on dialysis. *Arch Phys Med Rehabil*. 2020;101(2):227–233.
325. Grubbs V, Moss AH, Cohen LM, et al. A palliative approach to dialysis care: a patient-centered transition to the end of life. *Clin J Am Soc Nephrol*. 2014;9(12):2203–2209.
328. Vandecasteele SJ, Kurella Tamura M. A patient-centered vision of care for ESRD: dialysis as a bridging treatment or as a final destination? *J Am Soc Nephrol*. 2014;25(8):1647–1651.
333. Inker LA, Schmid CH, Tighiouart H, et al. Estimating glomerular filtration rate from serum creatinine and cystatin C. *N Engl J Med*. 2012;367(1):20–29.
353. Yazici AE, Erdem P, Erdem A, et al. Efficacy and tolerability of escitalopram in depressed patients with end stage renal disease: an open placebo-controlled study. *Klin Psikofarmakol B*. 2012;22(1):23–30.
354. Levy NB, Blumenfeld M, Beasley Jr CM, et al. Fluoxetine in depressed patients with renal failure and in depressed patients with normal kidney function. *Gen Hosp Psychiatry*. 1996;18(1):8–13.
355. Blumenfeld M, Levy NB, Spinowitz B, et al. Fluoxetine in depressed patients on dialysis. *Int J Psychiatry Med*. 1997;27(1):71–80.

REFERENCES

- Rule AD, Amer H, Cornell LD, Taler SJ, Cosio FG, Kremers WK, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152(9):561–567.
- Winearls CG, Glasscock RJ. Dissecting and refining the staging of chronic kidney disease. *Kidney Int.* 2009;75(10):1009–1014.
- Rule AD, Gussak HM, Pond GR, et al. Measured and estimated GFR in healthy potential kidney donors. *Am J Kidney Dis.* 2004;43(1):112–119.
- Glasscock RJ, Warnock DG, Delanaye P. The global burden of chronic kidney disease: estimates, variability and pitfalls. *Nat Rev Nephrol.* 2017;13(2):104–114.
- Coresh J. Update on the burden of CKD. *J Am Soc Nephrol.* 2017;28(4):1020–1022.
- Delanaye P, Glasscock RJ, Pottel H, Rule AD. An age-calibrated definition of chronic kidney disease: rationale and benefits. *Clin Biochem Rev.* 2016;37(1):17–26.
- Glasscock R, Delanaye P, El Nahas M. An age-calibrated classification of chronic kidney disease. *JAMA.* 2015;314(6):559–560.
- Stevens RJ, Evans J, Oke J, et al. Kidney age, not kidney disease. *CMAJ (Can Med Assoc J).* 2018;190(13):E389–E393.
- Delanaye P, Gaillard F, van der Weijden J, et al. Age-adapted percentiles of measured glomerular filtration in healthy individuals: extrapolation to living kidney donors over 65 years. *Clin Chem Lab Med.* 2022;60(3):401–407.
- Zoccali C, Plebani M. Extrapolated normative GFR data for living kidney donation. *Clin Chem Lab Med.* 2022;60(3):301–304.
- Andrassy KM. Comments on 'KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease'. *Kidney Int.* 2013;84(3):622–623.
- Beridze G, Vetrano DL, Marengoni A, Dai L, Carrero JJ, Calderon-Larranaga A. Concordance and discrepancies among 5 creatinine-based equations for assessing estimated glomerular filtration rate in older adults. *JAMA Netw Open.* 2023;6(3):e234211.
- Carnevale V, Tinti MG. Use of eGFR in older adults with kidney disease. *N Engl J Med.* 2022;387(6):573–574.
- Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med.* 2009;150(9):604–612.
- Farrington K, Covic A, Aucella F, et al. Clinical Practice Guideline on management of older patients with chronic kidney disease stage 3b or higher (eGFR <45 mL/min/1.73 m²). *Nephrol Dial Transplant.* 2016;31(suppl 2):ii1–ii66.
- Delanaye P, Pottel H, Melsom T. Physiology of the aging kidney: we know where we are going, but we don't know how. *Clin J Am Soc Nephrol.* 2022.
- Pottel H, Hoste L, Yayo E, Delanaye P. Glomerular filtration rate in healthy living potential kidney donors: a meta-analysis supporting the construction of the full age spectrum equation. *Nephron.* 2017;135(2):105–119.
- Rule AD, Cornell LD, Poggio ED. Senile nephrosclerosis—does it explain the decline in glomerular filtration rate with aging. *Nephron Physiol.* 2011;119(Suppl. 1):p6–p11.
- Glasscock R, Denic A, Rule AD. When kidneys get old: an essay on nephro-geriatrics. *J Bras Nefrol.* 2017;39(1):59–64.
- Glasscock RJ, Rule AD. The implications of anatomical and functional changes of the aging kidney: with an emphasis on the glomeruli. *Kidney Int.* 2012;82(3):270–277.
- Arora P, Jalal K, Gupta A, Carter RL, Lohr JW. Progression of kidney disease in elderly stage 3 and 4 chronic kidney disease patients. *Int Urol Nephrol.* 2017;49(6):1033–1040.
- Schaeffner ES, Ebert N, Kuhlmann MK, et al. Age and the course of GFR in persons aged 70 and above. *Clin J Am Soc Nephrol.* 2022.
- O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol.* 2007;18(10):2758–2765.
- Conway B, Webster A, Ramsay G, et al. Predicting mortality and uptake of renal replacement therapy in patients with stage 4 chronic kidney disease. *Nephrol Dial Transplant.* 2009;24(6):1930–1937.
- Turin TC, James M, Ravani P, et al. Proteinuria and rate of change in kidney function in a community-based population. *J Am Soc Nephrol.* 2013;24(10):1661–1667.
- Moranne O, Couchoud C, Vigneau C, Investigators PS. Characteristics and treatment course of patients older than 75 years, reaching end-stage renal failure in France. The PSPA study. *J Gerontol A Biol Sci Med Sci.* 2012;67(12):1394–1399.
- Levey AS, Grams ME, Inker LA. Uses of GFR and albuminuria level in acute and chronic kidney disease. *N Engl J Med.* 2022;386(22):2120–2128.
- Ebert N, Jakob O, Gaedeke J, et al. Prevalence of reduced kidney function and albuminuria in older adults: the Berlin Initiative Study. *Nephrol Dial Transplant.* 2017;32(6):997–1005.
- Garg AX, Kiberd BA, Clark WF, Haynes RB, Clase CM. Albuminuria and renal insufficiency prevalence guides population screening: results from the NHANES III. *Kidney Int.* 2002;61(6):2165–2175.
- Dalrymple LS, Katz R, Kestenbaum B, et al. Chronic kidney disease and the risk of end-stage renal disease versus death. *J Gen Intern Med.* 2011;26(4):379–385.
- Hemmelgarn BR, Manns BJ, Lloyd A, et al. Relation between kidney function, proteinuria, and adverse outcomes. *JAMA.* 2010;303(5):423–429.
- Go AS, Chertow GM, Fan D, McCulloch CE, Hsu CY. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med.* 2004;351(13):1296–1305.
- Shlipak MG, Stehman-Breen C, Fried LF, et al. The presence of frailty in elderly persons with chronic renal insufficiency. *Am J Kidney Dis.* 2004;43(5):861–867.
- Stevens LA, Li S, Wang C, et al. Prevalence of CKD and comorbid illness in elderly patients in the United States: results from the Kidney Early Evaluation Program (KEEP). *Am J Kidney Dis.* 2010;55(3 Suppl 2):S23–S33.
- Fried LF, Shlipak MG, Crump C, et al. Renal insufficiency as a predictor of cardiovascular outcomes and mortality in elderly individuals. *J Am Coll Cardiol.* 2003;41(8):1364–1372.
- Manjunath G, Tighiouart H, Coresh J, et al. Level of kidney function as a risk factor for cardiovascular outcomes in the elderly. *Kidney Int.* 2003;63(3):1121–1129.
- Bansal N, Katz R, De Boer IH, et al. Development and validation of a model to predict 5-year risk of death without ESRD among older adults with CKD. *Clin J Am Soc Nephrol.* 2015;10(3):363–371.
- Johnson ES, Thorp ML, Yang X, Charansonney OL, Smith DH. Predicting renal replacement therapy and mortality in CKD. *Am J Kidney Dis.* 2007;50(4):559–565.
- Weiss JW, Platt RW, Thorp ML, et al. Predicting mortality in older adults with kidney disease: a pragmatic prediction model. *J Am Geriatr Soc.* 2015;63(3):508–515.
- Tangri N, Grams ME, Levey AS, et al. Multinational assessment of accuracy of equations for predicting risk of kidney failure: a meta-analysis. *JAMA.* 2016;315(2):164–174.
- Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA.* 2011;305(15):1553–1559.
- Grams ME, Brunskill NJ, Ballew SH, et al. The kidney failure risk equation: evaluation of novel input variables including eGFR estimated using the CKD-EPI 2021 equation in 59 cohorts. *J Am Soc Nephrol.* 2023;34(3):482–494.
- Crews DC, Scialla JJ, Liu J, et al. Predialysis health, dialysis timing, and outcomes among older United States adults. *J Am Soc Nephrol.* 2014;25(2):370–379.
- Ishani A, Xue JL, Himmelfarb J, et al. Acute kidney injury increases risk of ESRD among elderly. *J Am Soc Nephrol.* 2009;20(1):223–228.
- Rosansky SJ, Schell J, Shega J, et al. Treatment decisions for older adults with advanced chronic kidney disease. *BMC Nephrol.* 2017;18(1):200.
- Kuo HK, Scandrett KG, Dave J, Mitchell SL. The influence of outpatient comprehensive geriatric assessment on survival: a meta-analysis. *Arch Gerontol Geriatr.* 2004;39(3):245–254.
- Stuck AE, Siu AL, Wieland GD, Adams J, Rubenstein LZ. Comprehensive geriatric assessment: a meta-analysis of controlled trials. *Lancet.* 1993;342(8878):1032–1036.
- Ellis G, Whitehead MA, Robinson D, O'Neill D, Langhorne P. Comprehensive geriatric assessment for older adults admitted to hospital: meta-analysis of randomised controlled trials. *BMJ.* 2011;343:d6553.
- Hall RK, Haines C, Gorbaten SM, et al. Incorporating geriatric assessment into a nephrology clinic: preliminary data from two models of care. *J Am Geriatr Soc.* 2016;64(10):2154–2158.

50. Racic M, Petkovic N, Bogicevic K, et al. Comprehensive geriatric assessment: comparison of elderly hemodialysis patients and primary care patients. *Ren Fail*. 2015;37(7):1126–1131.
51. Aucella F, Stoico L, Cicchella A, et al. Comprehensive geriatric assessment in the hemodialysis elderly population. *J Nephrol*. 2012;25(Suppl 19):S85–S89.
52. Parlevliet JL, Buurman BM, Pannekeet MM, et al. Systematic comprehensive geriatric assessment in elderly patients on chronic dialysis: a cross-sectional comparative and feasibility study. *BMC Nephrol*. 2012;13:30.
53. Rosner M, Abdel-Rahman E, Williams ME, Nephrology ASNA-GoG. Geriatric nephrology: responding to a growing challenge. *Clin J Am Soc Nephrol*. 2010;5(5):936–942.
54. Morley JE, Vellas B, van Kan GA, et al. Frailty consensus: a call to action. *J Am Med Dir Assoc*. 2013;14(6):392–397.
55. Gobbens RJ, Luijckx KG, Wijnen-Sponselee MT, Schols JM. In search of an integral conceptual definition of frailty: opinions of experts. *J Am Med Dir Assoc*. 2010;11(5):338–343.
56. Sutton JL, Gould RL, Daley S, et al. Psychometric properties of multicomponent tools designed to assess frailty in older adults: a systematic review. *BMC Geriatr*. 2016;16:55.
57. Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci*. 2001;56(3):M146–M156.
58. Bowling CB, Booth 3rd JN, Safford MM, et al. Nondisease-specific problems and all-cause mortality in the REasons for geographic and racial differences in stroke study. *J Am Geriatr Soc*. 2013;61(5):739–746.
59. Bowling CB, Muntner P, Sawyer P, et al. Community mobility among older adults with reduced kidney function: a study of life-space. *Am J Kidney Dis*. 2014;63(3):429–436.
60. Johansen KL. The frail dialysis population: a growing burden for the dialysis community. *Blood Purif*. 2015;40(4):288–292.
61. Musso CG, Jauregui JR, Macias Nunez JF. Frailty phenotype and chronic kidney disease: a review of the literature. *Int Urol Nephrol*. 2015;47(11):1801–1807.
62. Bao Y, Dalrymple L, Chertow GM, Kaysen GA, Johansen KL. Frailty, dialysis initiation, and mortality in end-stage renal disease. *Arch Intern Med*. 2012;172(14):1071–1077.
63. Delgado C, Grimes BA, Glidden DV, Shlipak M, Sarnak MJ, Johansen KL. Association of Frailty based on self-reported physical function with directly measured kidney function and mortality. *BMC Nephrol*. 2015;16:203.
64. Chowdhury R, Peel NM, Krosch M, Hubbard RE. Frailty and chronic kidney disease: a systematic review. *Arch Gerontol Geriatr*. 2017;68:135–142.
65. Roshanravan B, Khatri M, Robinson-Cohen C, et al. A prospective study of frailty in nephrology-referred patients with CKD. *Am J Kidney Dis*. 2012;60(6):912–921.
66. Collard RM, Boter H, Schoevers RA, Oude Voshaar RC. Prevalence of frailty in community-dwelling older persons: a systematic review. *J Am Geriatr Soc*. 2012;60(8):1487–1492.
67. Johansen KL, Delgado C, Bao Y, Kurella Tamura M. Frailty and dialysis initiation. *Semin Dial*. 2013;26(6):690–696.
68. Kojima G. Prevalence of frailty in end-stage renal disease: a systematic review and meta-analysis. *Int Urol Nephrol*. 2017;49(11):1989–1997.
69. Johansen KL, Chertow GM, Jin C, Kutner NG. Significance of frailty among dialysis patients. *J Am Soc Nephrol*. 2007;18(11):2960–2967.
70. Lee SY, Yang DH, Hwang E, et al. The prevalence, association, and clinical outcomes of frailty in maintenance dialysis patients. *J Ren Nutr*. 2017;27(2):106–112.
71. Kim JC, Kalantar-Zadeh K, Kopple JD. Frailty and protein-energy wasting in elderly patients with end stage kidney disease. *J Am Soc Nephrol*. 2013;24(3):337–351.
72. Johansen KL, Dalrymple LS, Delgado C, et al. Factors associated with frailty and its trajectory among patients on hemodialysis. *Clin J Am Soc Nephrol*. 2017;12(7):1100–1108.
73. Mitnitski A, Song X, Rockwood K. Trajectories of changes over twelve years in the health status of Canadians from late middle age. *Exp Gerontol*. 2012;47(12):893–899.
74. Gill TM, Gahbauer EA, Allore HG, Han L. Transitions between frailty states among community-living older persons. *Arch Intern Med*. 2006;166(4):418–423.
75. Bowling CB, Sawyer P, Campbell RC, Ahmed A, Allman RM. Impact of chronic kidney disease on activities of daily living in community-dwelling older adults. *J Gerontol A Biol Sci Med Sci*. 2011;66(6):689–694.
76. Kurella Tamura M, Covinsky KE, Chertow GM, Yaffe K, Landefeld CS, McCulloch CE. Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med*. 2009;361(16):1539–1547.
77. Jassal SV, Chiu E, Hladunewich M. Loss of independence in patients starting dialysis at 80 years of age or older. *N Engl J Med*. 2009;361(16):1612–1613.
78. Lattanzio F, Corsonello A, Abbatecola AM, et al. Relationship between renal function and physical performance in elderly hospitalized patients. *Rejuvenation Res*. 2012;15(6):545–552.
79. Rampersad C, Darcel J, Harasemiw O, et al. Change in physical activity and function in patients with baseline advanced nondialysis CKD. *Clin J Am Soc Nephrol*. 2021;16(12):1805–1812.
80. Goto NA, van Loon IN, Boereboom FTJ, et al. Association of initiation of maintenance dialysis with functional status and caregiver burden. *Clin J Am Soc Nephrol*. 2019;14(7):1039–1047.
81. Jassal SV, Karaboyas A, Comment LA, et al. Functional dependence and mortality in the international dialysis outcomes and practice patterns study (DOPPS). *Am J Kidney Dis*. 2016;67(2):283–292.
82. Chaudhry SI, McAvay G, Ning Y, Allore HG, Newman AB, Gill TM. Geriatric impairments and disability: the cardiovascular health study. *J Am Geriatr Soc*. 2010;58(9):1686–1692.
83. Tennankore K, Zhao J, Karaboyas A, et al. The association of functional status with mortality and dialysis modality change: results from the peritoneal dialysis outcomes and practice patterns study (PDOPPS). *Perit Dial Int*. 2019;39(2):103–111.
84. Covinsky KE, Palmer RM, Fortinsky RH, et al. Loss of independence in activities of daily living in older adults hospitalized with medical illnesses: increased vulnerability with age. *J Am Geriatr Soc*. 2003;51(4):451–458.
85. Sands LP, Xu H, Craig BA, Eng C, Covinsky KE. Predicting change in functional status over quarterly intervals for older adults enrolled in the PACE community-based long-term care program. *Aging Clin Exp Res*. 2008;20(5):419–427.
86. Carey EC, Covinsky KE, Lui LY, Eng C, Sands LP, Walter LC. Prediction of mortality in community-living frail elderly people with long-term care needs. *J Am Geriatr Soc*. 2008;56(1):68–75.
87. Carey EC, Walter LC, Lindquist K, Covinsky KE. Development and validation of a functional morbidity index to predict mortality in community-dwelling elders. *J Gen Intern Med*. 2004;19(10):1027–1033.
88. Gill TM, Kurland B. The burden and patterns of disability in activities of daily living among community-living older persons. *J Gerontol A Biol Sci Med Sci*. 2003;58(1):70–75.
89. Onder G, Penninx BW, Ferrucci L, Fried LP, Guralnik JM, Pahor M. Measures of physical performance and risk for progressive and catastrophic disability: results from the Women's Health and Aging Study. *J Gerontol A Biol Sci Med Sci*. 2005;60(1):74–79.
90. Dussaux E, Albano L, Fafin C, et al. A simple clinical tool to inform the decision-making process to refer elderly incident dialysis patients for kidney transplant evaluation. *Kidney Int*. 2015;88(1):121–129.
91. Arai Y, Kanda E, Kikuchi H, et al. Decreased mobility after starting dialysis is an independent risk factor for short-term mortality after initiation of dialysis. *Nephrology (Carlton)*. 2014;19(4):227–233.
92. Couchoud C, Labeeuw M, Moranne O, et al. A clinical score to predict 6-month prognosis in elderly patients starting dialysis for end-stage renal disease. *Nephrol Dial Transplant*. 2009;24(5):1553–1561.
93. Couchoud CG, Beuscart JB, Aldigier JC, Brunet PJ, Moranne OP, registry R. Development of a risk stratification algorithm to improve patient-centered care and decision making for incident elderly patients with end-stage renal disease. *Kidney Int*. 2015;88(5):1178–1186.
94. Glaudet F, Hottelart C, Allard J, et al. The clinical status and survival in elderly dialysis: example of the oldest region of France. *BMC Nephrol*. 2013;14:131.
95. Thamer M, Kaufman JS, Zhang Y, Zhang Q, Cotter DJ, Bang H. Predicting early death among elderly dialysis patients: development and validation of a risk score to assist shared decision making for dialysis initiation. *Am J Kidney Dis*. 2015;66(6):1024–1032.

96. Kurella M, Covinsky KE, Collins AJ, Chertow GM. Octogenarians and nonagenarians starting dialysis in the United States. *Ann Intern Med*. 2007;146(3):177–183.
97. Belkin MD, Doerfler RM, Wagner LA, Zhan M, Fink JC. Associations of performance-based functional assessments and adverse outcomes in CKD. *Kidney*. 2021;2(4):629–638.
98. Inouye SK, Studenski S, Tinetti ME, Kuchel GA. Geriatric syndromes: clinical, research, and policy implications of a core geriatric concept. *J Am Geriatr Soc*. 2007;55(5):780–791.
99. Bowling CB, Bromfield SG, Colantonio LD, et al. Association of reduced eGFR and albuminuria with serious fall injuries among older adults. *Clin J Am Soc Nephrol*. 2016;11(7):1236–1243.
100. Farragher J, Chiu E, Ulutas O, Tomlinson G, Cook WL, Jassal SV. Accidental falls and risk of mortality among older adults on chronic peritoneal dialysis. *Clin J Am Soc Nephrol*. 2014;9(7):1248–1253.
101. Tinetti ME, Williams CS. Falls, injuries due to falls, and the risk of admission to a nursing home. *N Engl J Med*. 1997;337(18):1279–1284.
102. Sattin RW, Lambert Huber DA, DeVito CA, et al. The incidence of fall injury events among the elderly in a defined population. *Am J Epidemiol*. 1990;131(6):1028–1037.
103. Cook WL, Tomlinson G, Donaldson M, et al. Falls and fall-related injuries in older dialysis patients. *Clin J Am Soc Nephrol*. 2006;1(6):1197–1204.
104. Bergen G, Stevens MR, Burns ER. Falls and fall injuries among adults aged >=65 Years - United States, 2014. *MMWR Morb Mortal Wkly Rep*. 2016;65(37):993–998.
105. Sterling DA, O'Connor JA, Bonadies J. Geriatric falls: injury severity is high and disproportionate to mechanism. *J Trauma*. 2001;50(1):116–119.
106. Alexander BH, Rivara FP, Wolf ME. The cost and frequency of hospitalization for fall-related injuries in older adults. *Am J Public Health*. 1992;82(7):1020–1023.
107. Cumming RG, Salkeld G, Thomas M, Szonyi G. Prospective study of the impact of fear of falling on activities of daily living, SF-36 scores, and nursing home admission. *J Gerontol A Biol Sci Med Sci*. 2000;55(5):M299–305.
108. Farragher J, Rajan T, Chiu E, et al. Equivalent fall risk in elderly patients on hemodialysis and peritoneal dialysis. *Perit Dial Int*. 2016;36(1):67–70.
109. McAdams-DeMarco MA, Suresh S, Law A, et al. Frailty and falls among adult patients undergoing chronic hemodialysis: a prospective cohort study. *BMC Nephrol*. 2013;14:224.
110. Plantinga LC, Patzer RE, Franch HA, Bowling CB. Serious fall injuries before and after initiation of hemodialysis among older ESRD patients in the United States: a retrospective cohort study. *Am J Kidney Dis*. 2017;70(1):76–83.
111. Tran J, Ayers E, Verghese J, Abramowitz MK. Gait abnormalities and the risk of falls in CKD. *Clin J Am Soc Nephrol*. 2019;14(7):983–993.
112. Palmer S, Vecchio M, Craig JC, et al. Prevalence of depression in chronic kidney disease: systematic review and meta-analysis of observational studies. *Kidney Int*. 2013;84(1):179–191.
113. Shirazian S, Grant CD, Aina O, Mattana J, Khorassani F, Ricardo AC. Depression in chronic kidney disease and end-stage renal disease: similarities and differences in diagnosis, Epidemiology, and management. *Kidney Int Rep*. 2017;2(1):94–107.
114. Cukor D, Fruchter Y, Ver Halen N, Naidoo S, Patel A, Saggi SJ. A preliminary investigation of depression and kidney functioning in patients with chronic kidney disease. *Nephron Clin Pract*. 2012;122(3–4):139–145.
115. Lee YJ, Kim MS, Cho S, Kim SR. Association of depression and anxiety with reduced quality of life in patients with predialysis chronic kidney disease. *Int J Clin Pract*. 2013;67(4):363–368.
116. Lopes GB, Matos CM, Leite EB, et al. Depression as a potential explanation for gender differences in health-related quality of life among patients on maintenance hemodialysis. *Nephron Clin Pract*. 2010;115(1):c35–c40.
117. Fischer MJ, Kimmel PL, Greene T, et al. Sociodemographic factors contribute to the depressive affect among African Americans with chronic kidney disease. *Kidney Int*. 2010;77(11):1010–1019.
118. Kurella M, Kimmel PL, Young BS, Chertow GM. Suicide in the United States end-stage renal disease program. *J Am Soc Nephrol*. 2005;16(3):774–781.
119. Weisbord SD, Mor MK, Sevic MA, et al. Associations of depressive symptoms and pain with dialysis adherence, health resource utilization, and mortality in patients receiving chronic hemodialysis. *Clin J Am Soc Nephrol*. 2014;9(9):1594–1602.
120. Hedayati SS, Grambow SC, Szczech LA, Stechuchak KM, Allen AS, Bosworth HB. Physician-diagnosed depression as a correlate of hospitalizations in patients receiving long-term hemodialysis. *Am J Kidney Dis*. 2005;46(4):642–649.
121. Hedayati SS, Minhajuddin AT, Afshar M, Toto RD, Trivedi MH, Rush AJ. Association between major depressive episodes in patients with chronic kidney disease and initiation of dialysis, hospitalization, or death. *JAMA*. 2010;303(19):1946–1953.
122. Palmer SC, Vecchio M, Craig JC, et al. Association between depression and death in people with CKD: a meta-analysis of cohort studies. *Am J Kidney Dis*. 2013;62(3):493–505.
123. Farrokhi F, Abedi N, Beyene J, Kurdyak P, Jassal SV. Association between depression and mortality in patients receiving long-term dialysis: a systematic review and meta-analysis. *Am J Kidney Dis*. 2014;63(4):623–635.
124. Chiang HH, Guo HR, Livneh H, Lu MC, Yen ML, Tsai TY. Increased risk of progression to dialysis or death in CKD patients with depressive symptoms: a prospective 3-year follow-up cohort study. *J Psychosom Res*. 2015;79(3):228–232.
125. Chen HY, Chiang CK, Wang HH, et al. Cognitive-behavioral therapy for sleep disturbance in patients undergoing peritoneal dialysis: a pilot randomized controlled trial. *Am J Kidney Dis*. 2008;52(2):314–323.
126. Duarte PS, Miyazaki MC, Blay SL, Sesso R. Cognitive-behavioral group therapy is an effective treatment for major depression in hemodialysis patients. *Kidney Int*. 2009;76(4):414–421.
127. Cukor D, Ver Halen N, Asher DR, et al. Psychosocial intervention improves depression, quality of life, and fluid adherence in hemodialysis. *J Am Soc Nephrol*. 2014;25(1):196–206.
128. Regier DA, Narrow WE, Kuhl EA, Kupfer DJ, Association AP. *The Conceptual Evolution of DSM-5*. American Psychiatric Pub; 2010.
129. Pena-Polanco JE, Mor MK, Tohme FA, Fine MJ, Palevsky PM, Weisbord SD. Acceptance of antidepressant treatment by patients on hemodialysis and their renal providers. *Clin J Am Soc Nephrol*. 2017;12(2):298–303.
130. Lopes AA, Albert JM, Young EW, et al. Screening for depression in hemodialysis patients: associations with diagnosis, treatment, and outcomes in the DOPPS. *Kidney Int*. 2004;66(5):2047–2053.
131. Wuerth D, Finkelstein SH, Ciarcia J, Peterson R, Klinger AS, Finkelstein FO. Identification and treatment of depression in a cohort of patients maintained on chronic peritoneal dialysis. *Am J Kidney Dis*. 2001;37(5):1011–1017.
132. Fischer MJ, Xie D, Jordan N, et al. Factors associated with depressive symptoms and use of antidepressant medications among participants in the Chronic Renal Insufficiency Cohort (CRIC) and Hispanic-CRIC Studies. *Am J Kidney Dis*. 2012;60(1):27–38.
133. Farrokhi F, Beanlands H, Logan A, Kurdyak P, Jassal SV. Patient-perceived barriers to a screening program for depression: a patient opinion survey of hemodialysis patients. *Clinical kidney journal*. 2017;10(6):830–837.
134. Nagler E, Webster AC, Vanholder R, Zoccali C. Antidepressants for depression in stage 3–5 chronic kidney disease: a systematic review of pharmacokinetics, efficacy and safety with recommendations by European Renal Best Practice (ERBP). *Nephrol Dial Transplant*. 2012;27(10):3736–3745.
135. Palmer SC, Natale P, Ruospo M, et al. Antidepressants for treating depression in adults with end-stage kidney disease treated with dialysis. *Cochrane Database Syst Rev*. 2016;2016(5):CD004541.
136. Cohen SD, Norris L, Acquaviva K, Peterson RA, Kimmel PL. Screening, diagnosis, and treatment of depression in patients with end-stage renal disease. *Clin J Am Soc Nephrol*. 2007;2(6):1332–1342.
137. Hedayati SS, Yalamanchili V, Finkelstein FO. A practical approach to the treatment of depression in patients with chronic kidney disease and end-stage renal disease. *Kidney Int*. 2012;81(3):247–255.
138. Birkinshaw H, Friedrich CM, Cole P, et al. Antidepressants for pain management in adults with chronic pain: a network meta-analysis. *Cochrane Database Syst Rev*. 2023;5(5):CD014682.
139. Lobo ED, Heathman M, Kuan HY, et al. Effects of varying degrees of renal impairment on the pharmacokinetics of duloxetine:

- analysis of a single-dose phase I study and pooled steady-state data from phase II/III trials. *Clin Pharmacokinet*. 2010;49(5):311–321.
140. Viktil KK, Blix HS, Moger TA, Reikvam A. Polypharmacy as commonly defined is an indicator of limited value in the assessment of drug-related problems. *Br J Clin Pharmacol*. 2007;63(2):187–195.
 141. Kantor ED, Rehm CD, Haas JS, Chan AT, Giovannucci EL. Trends in prescription drug use among adults in the United States from 1999–2012. *JAMA*. 2015;314(17):1818–1831.
 142. Maher RL, Hanlon J, Hajjar ER. Clinical consequences of polypharmacy in elderly. *Expert Opin Drug Saf*. 2014;13(1):57–65.
 143. Gerardi S, Sperlea D, Levy SO, et al. Implementation of targeted deprescribing of potentially inappropriate medications in patients on hemodialysis. *Am J Health Syst Pharm*. 2022;79(Suppl 4):S128–S135.
 144. McLean AJ, Le Couteur DG. Aging biology and geriatric clinical pharmacology. *Pharmacol Rev*. 2004;56(2):163–184.
 145. Lassiter J, Bennett WM, Olyaei AJ. Drug dosing in elderly patients with chronic kidney disease. *Clin Geriatr Med*. 2013;29(3):657–705.
 146. Van Spall HG, Toren A, Kiss A, Fowler RA. Eligibility criteria of randomized controlled trials published in high-impact general medical journals: a systematic sampling review. *JAMA*. 2007;297(11):1233–1240.
 147. Hubbard RE, O'Mahony MS, Woodhouse KW. Medication prescribing in frail older people. *Eur J Clin Pharmacol*. 2013;69(3):319–326.
 148. Rifkin DE, Winkelmayer WC. Medication issues in older individuals with CKD. *Adv Chronic Kidney Dis*. 2010;17(4):320–328.
 149. Verbeeck RK, Musumba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*. 2009;65(8):757–773.
 150. Novick TK, Surapaneni A, Shin JI, et al. Associations of opioid prescriptions with death and hospitalization across the spectrum of estimated GFR. *Clin J Am Soc Nephrol*. 2019;14(11):1581–1589.
 151. Vangala C, Niu J, Montez-Rath ME, et al. Hip fracture risk among hemodialysis-dependent patients prescribed opioids and gabapentinoids. *J Am Soc Nephrol*. 2020;31(6):1325–1334.
 152. Chauvin KJ, Blake PG, Garg AX, et al. Baclofen has a risk of encephalopathy in older adults receiving dialysis. *Kidney Int*. 2020;98(4):979–988.
 153. American Geriatrics Society Beers Criteria Update Expert P. American Geriatrics Society 2023 updated AGS Beers Criteria(R) for potentially inappropriate medication use in older adults. *J Am Geriatr Soc*. 2023.
 154. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med*. 2023.
 155. Santin FG, Bigogno FG, Dias Rodrigues JC, Cuppari L, Avesani CM. Concurrent and predictive validity of composite methods to assess nutritional status in older adults on hemodialysis. *J Ren Nutr*. 2016;26(1):18–25.
 156. Carrero JJ, Stenvinkel P, Cuppari L, et al. Etiology of the protein-energy wasting syndrome in chronic kidney disease: a consensus statement from the International Society of Renal Nutrition and Metabolism (ISRNM). *J Ren Nutr*. 2013;23(2):77–90.
 157. Obi Y, Qader H, Kovesdy CP, Kalantar-Zadeh K. Latest consensus and update on protein-energy wasting in chronic kidney disease. *Curr Opin Clin Nutr Metab Care*. 2015;18(3):254–262.
 158. Johansson L, Fouque D, Bellizzi V, et al. As we grow old: nutritional considerations for older patients on dialysis. *Nephrol Dial Transplant*. 2017;32(7):1127–1136.
 159. Bossola M, Tazza L, Giungi S, Luciani G. Anorexia in hemodialysis patients: an update. *Kidney Int*. 2006;70(3):417–422.
 160. Carrero JJ. Identification of patients with eating disorders: clinical and biochemical signs of appetite loss in dialysis patients. *J Ren Nutr*. 2009;19(1):10–15.
 161. Sekiguchi T, Kabayama M, Ryuno H, et al. Association between protein intake and changes in renal function among Japanese community-dwelling older people: the SONIC study. *Geriatr Gerontol Int*. 2022;22(4):286–291.
 162. Molnar MZ, Sumida K, Gaipov A, et al. Pre-ESRD dementia and post-ESRD mortality in a large cohort of incident dialysis patients. *Dement Geriatr Cogn Disord*. 2017;43(5–6):281–293.
 163. Sehgal AR, Grey SF, DeOreo PB, Whitehouse PJ. Prevalence, recognition, and implications of mental impairment among hemodialysis patients. *Am J Kidney Dis*. 1997;30(1):41–49.
 164. Griva K, Stygal J, Hankins M, Davenport A, Harrison M, Newman SP. Cognitive impairment and 7-year mortality in dialysis patients. *Am J Kidney Dis*. 2010;56(4):693–703.
 165. Anazodo UC, Wong DY, Théberge J, et al. Hemodialysis-related acute brain injury demonstrated by application of intradialytic magnetic resonance imaging and spectroscopy. *J Am Soc Nephrol*. 2023;34(6):1090–1104.
 166. Jassal SV, Farragher JF. MoCA: turn your mind to it. *J Am Soc Nephrol*. 2020;31(4):672–673.
 167. Kurella Tamura M, Vittinghoff E, Hsu CY, et al. Loss of executive function after dialysis initiation in adults with chronic kidney disease. *Kidney Int*. 2017;91(4):948–953.
 168. Weiss J, Freeman M, Low A, et al. Benefits and harms of intensive blood pressure treatment in adults aged 60 Years or older: a systematic review and meta-analysis. *Ann Intern Med*. 2017;166(6):419–429.
 169. Textor SC, Schwartz GL. *BP Targets in CKD, Mortality, and SPRINT: What Have We Learned?*. Am Soc Nephrol; 2017:2561–2563.
 170. Group SR, Wright Jr JT, Williamson JD, et al. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med*. 2015;373(22):2103–2116.
 171. Williamson JD, Supiano MA, Applegate WB, et al. Intensive vs standard blood pressure control and cardiovascular disease outcomes in adults aged ≥ 75 Years: a randomized clinical trial. *JAMA*. 2016;315(24):2673–2682.
 172. Covic A, Apetrii M, Goldsmith D, Kanbay M. SPRINT: the study nephrologists might take with a grain of salt. *J Clin Hypertens (Greenwich)*. 2016;18(11):1185–1188.
 173. Xie X, Atkins E, Lv J, et al. Effects of intensive blood pressure lowering on cardiovascular and renal outcomes: updated systematic review and meta-analysis. *Lancet*. 2016;387(10017):435–443.
 174. Benetos A, Labat C, Rossignol P, et al. Treatment with multiple blood pressure medications, achieved blood pressure, and mortality in older nursing home residents: the PARTAGE study. *JAMA Intern Med*. 2015;175(6):989–995.
 175. Zhang W, Zhang S, Deng Y, et al. Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med*. 2021;385(14):1268–1279.
 176. Tonelli M, Muntner P, Lloyd A, et al. Risk of coronary events in people with chronic kidney disease compared with those with diabetes: a population-level cohort study. *Lancet*. 2012;380(9844):807–814.
 177. Wanner C, Tonelli M. Kidney disease: improving global outcomes lipid guideline development work group M. KDIGO clinical practice guideline for lipid management in CKD: summary of recommendation statements and clinical approach to the patient. *Kidney Int*. 2014;85(6):1303–1309.
 178. Baigent C, Landray MJ, Reith C, et al. The effects of lowering LDL cholesterol with simvastatin plus ezetimibe in patients with chronic kidney disease (Study of Heart and Renal Protection): a randomised placebo-controlled trial. *Lancet*. 2011;377(9784):2181–2192.
 179. Butler CR, O'Hare AM. Considerations in applying the results of randomized controlled clinical trials to the care of older adults with kidney disease in the clinical setting: the SHARP trial. *Adv Chronic Kidney Dis*. 2016;23(1):29–35.
 180. Arnett DK, Blumenthal RS, Albert MA, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: executive summary: a report of the American College of Cardiology/American heart association task force on clinical practice guidelines. *Circulation*. 2019;140(11):e563–e595.
 181. Hawley CE, Roefaro J, Forman DE, Orkaby AR. Statins for primary prevention in those aged 70 years and older: a critical review of recent cholesterol guidelines. *Drugs Aging*. 2019;36(8):687–699.
 182. Ballantyne CM, Varughese MG, Abushamat LA. Lipid-lowering therapy in the elderly: are current guidelines a sign of ageism in medical care? *J Am Coll Cardiol*. 2023;81(14):1350–1352.
 183. Wanner C, Krane V, Marz W, et al. Atorvastatin in patients with type 2 diabetes mellitus undergoing hemodialysis. *N Engl J Med*. 2005;353(3):238–248.
 184. Fellstrom BC, Jardine AG, Schmieder RE, et al. Rosuvastatin and cardiovascular events in patients undergoing hemodialysis. *N Engl J Med*. 2009;360(14):1395–1407.
 185. Upadhyay A, Earley A, Lamont JL, Haynes S, Wanner C, Balk EM. Lipid-lowering therapy in persons with chronic kidney

- disease: a systematic review and meta-analysis. *Ann Intern Med.* 2012;157(4):251–262.
186. Palmer SC, Craig JC, Navaneethan SD, Tonelli M, Pellegrini F, Strippoli GF. Benefits and harms of statin therapy for persons with chronic kidney disease: a systematic review and meta-analysis. *Ann Intern Med.* 2012;157(4):263–275.
 187. Hou W, Lv J, Perkovic V, et al. Effect of statin therapy on cardiovascular and renal outcomes in patients with chronic kidney disease: a systematic review and meta-analysis. *Eur Heart J.* 2013;34(24):1807–1817.
 188. Messow C, Isles C. Meta-analysis of statins in chronic kidney disease: who benefits? *QJM.* 2017;110(8):493–500.
 189. Holdaas H, Holme I, Schmieder RE, et al. Rosuvastatin in diabetic hemodialysis patients. *J Am Soc Nephrol.* 2011;22(7):1335–1341.
 190. De Vriese AS. Should statins Be banned from dialysis? *J Am Soc Nephrol.* 2017;28(6):1675–1676.
 191. Pearson GJ, Thanassoulis G, Anderson TJ, et al. 2021 Canadian cardiovascular society guidelines for the management of dyslipidemia for the prevention of cardiovascular disease in adults. *Can J Cardiol.* 2021;37(8):1129–1150.
 192. Piccini JP, Hammill BG, Sinner MF, et al. Incidence and prevalence of atrial fibrillation and associated mortality among Medicare beneficiaries, 1993–2007. *Circ Cardiovasc Qual Outcomes.* 2012;5(1):85–93.
 193. Goldstein BA, Arce CM, Hlatky MA, Turakhia M, Setoguchi S, Winkelmayer WC. Trends in the incidence of atrial fibrillation in older patients initiating dialysis in the United States. *Circulation.* 2012;126(19):2293–2301.
 194. Winkelmayer WC, Patrick AR, Liu J, Brookhart MA, Setoguchi S. The increasing prevalence of atrial fibrillation among hemodialysis patients. *J Am Soc Nephrol.* 2011;22(2):349–357.
 195. Welanders F, Renlund H, Dimeny E, Holmberg H, Sjalander A. Direct oral anticoagulants versus warfarin in patients with non-valvular atrial fibrillation and CKD G3–G5D. *Clin Kidney J.* 2023;16(5):835–844.
 196. Bansal N, Zelnick LR, An J, et al. Association of kidney function with risk of adverse effects of therapies for atrial fibrillation. *Kidney Int Rep.* 2023;8(3):606–618.
 197. Wanner C, Herzog CA, Turakhia MP, Conference Steering C. Chronic kidney disease and arrhythmias: highlights from a kidney disease: improving global outcomes (KDIGO) controversies conference. *Kidney Int.* 2018;94(2):231–234.
 198. Chan KE, Edelman ER, Wenger JB, Thadhani RI, Maddux FW. Dabigatran and rivaroxaban use in atrial fibrillation patients on hemodialysis. *Circulation.* 2015;131(11):972–979.
 199. Chan KE, Giugliano RP, Patel MR, et al. Nonvitamin K anticoagulant agents in patients with advanced chronic kidney disease or on dialysis with AF. *J Am Coll Cardiol.* 2016;67(24):2888–2899.
 200. Dyer SM, Crotty M, Fairhall N, et al. A critical review of the long-term disability outcomes following hip fracture. *BMC Geriatr.* 2016;16(1):158.
 201. Shen Y, Huang X, Wu J, et al. The global burden of osteoporosis, low bone mass, and its related fracture in 204 countries and territories, 1990–2019. *Front Endocrinol (Lausanne).* 2022;13:882241.
 202. Sidibé A, Auguste D, Desbiens LC, et al. Fracture risk in dialysis and kidney transplanted patients: a systematic review. *JBM Rep.* 2019;3(1):45–55.
 203. Coco M, Rush H. Increased incidence of hip fractures in dialysis patients with low serum parathyroid hormone. *Am J Kidney Dis.* 2000;36(6):1115–1121.
 204. Wilson LM, Rebholz CM, Jirru E, et al. Benefits and harms of osteoporosis medications in patients with chronic kidney disease: a systematic review and meta-analysis. *Ann Intern Med.* 2017;166(9):649–658.
 205. Evenepoel P, Cunningham J, Ferrari S, et al. Diagnosis and management of osteoporosis in chronic kidney disease stages 4 to 5D: a call for a shift from nihilism to pragmatism. *Osteoporos Int.* 2021;32(12):2397–2405.
 206. Khairallah P, Nickolas TL. Management of osteoporosis in CKD. *Clin J Am Soc Nephrol.* 2018;13(6):962–969.
 207. Ketteler M, Block GA, Evenepoel P, et al. Executive summary of the 2017 KDIGO chronic kidney disease-mineral and bone disorder (CKD-MBD) guideline update: what's changed and why it matters. *Kidney Int.* 2017;92(1):26–36.
 208. Toussaint ND, Elder GJ, Kerr PG. Bisphosphonates in chronic kidney disease; balancing potential benefits and adverse effects on bone and soft tissue. *Clin J Am Soc Nephrol.* 2009;4(1):221–233.
 209. Moist LM, Lok CE, Vachharajani TJ, et al. Optimal hemodialysis vascular access in the elderly patient. *Semin Dial.* 2012;25(6):640–648.
 210. O'Hare AM. Vascular access for hemodialysis in older adults: a “patient first” approach. *J Am Soc Nephrol.* 2013;24(8):1187–1190.
 211. Wish JB. Catheter last, fistula not-so-first. *J Am Soc Nephrol.* 2015;26(1):5–7.
 212. Al-Balas A, Lee T, Young CJ, Kepes JA, Barker-Finkel J, Allon M. The clinical and economic effect of vascular access selection in patients initiating hemodialysis with a catheter. *J Am Soc Nephrol.* 2017;28(12):3679–3687.
 213. Brown RS, Patibandla BK, Goldfarb-Rumyantzev AS. The survival benefit of “fistula first, catheter last” in hemodialysis is primarily due to patient factors. *J Am Soc Nephrol.* 2017;28(2):645–652.
 214. Quinn RR, Oliver MJ, Devoe D, et al. The effect of predialysis fistula attempt on risk of all-cause and access-related death. *J Am Soc Nephrol.* 2017;28(2):613–620.
 215. Rha B, See I, Dunham L, et al. Vital signs: health disparities in hemodialysis-associated *Staphylococcus aureus* bloodstream infections - United States, 2017–2020. *MMWR Morb Mortal Wkly Rep.* 2023;72(6):153–159.
 216. DeSilva RN, Patibandla BK, Vin Y, et al. Fistula first is not always the best strategy for the elderly. *J Am Soc Nephrol.* 2013;24(8):1297–1304.
 217. Drew DA, Lok CE, Cohen JT, Wagner M, Tangri N, Weiner DE. Vascular access choice in incident hemodialysis patients: a decision analysis. *J Am Soc Nephrol.* 2015;26(1):183–191.
 218. Vachharajani TJ, Moossavi S, Jordan JR, Vachharajani V, Freedman BI, Burkart JM. Re-Evaluating the fistula first initiative in octogenarians on hemodialysis. *Clin J Am Soc Nephrol.* 2011;6(7):1663–1667.
 219. Lok CE, Allon M, Moist L, Oliver MJ, Shah H, Zimmerman D. Risk equation determining unsuccessful cannulation events and failure to maturation in arteriovenous fistulas (REDUCE FTM I). *J Am Soc Nephrol.* 2006;17(11):3204–3212.
 220. Dember LM, Beck CJ, Allon M, et al. Effect of clopidogrel on early failure of arteriovenous fistulas for hemodialysis: a randomized controlled trial. *JAMA.* 2008;299(18):2164–2171.
 221. Woo K, Goldman DP, Romley JA. Early failure of dialysis access among the elderly in the era of fistula first. *Clin J Am Soc Nephrol.* 2015;10(10):1791–1798.
 222. Quinn RR, Lamping DL, Lok CE, et al. The Vascular Access Questionnaire: assessing patient-reported views of vascular access. *J Vasc Access.* 2008;9(2):122–128.
 223. Lee T, Barker J, Allon M. Needle infiltration of arteriovenous fistulae in hemodialysis: risk factors and consequences. *Am J Kidney Dis.* 2006;47(6):1020–1026.
 224. Rehfuess JP, Berceci SA, Barbey SM, et al. The spectrum of hand dysfunction after hemodialysis fistula placement. *Kidney Int Rep.* 2017;2(3):332–341.
 225. Maguire IC, Browne LD, Dawood M, et al. Differential impact of central venous catheters versus arteriovenous fistulae on quality of life among Irish haemodialysis patients. *Kidney360.* 2022;3(6):1065–1072.
 226. Viccelli AK, Lok CE. Hemodialysis vascular access in the elderly: getting it right. *Kidney Int.* 2019;95(1):38–49.
 227. O'Hare AM, Allon M, Kaufman JS. Whether and when to refer patients for predialysis AV fistula creation: complex decision making in the face of uncertainty. *Semin Dial.* 2010;23(5):452–455.
 228. Shingarev R, Maya ID, Barker-Finkel J, Allon M. Arteriovenous graft placement in predialysis patients: a potential catheter-sparing strategy. *Am J Kidney Dis.* 2011;58(2):243–247.
 229. van Walraven C, Manuel DG, Knoll G. Survival trends in ESRD patients compared with the general population in the United States. *Am J Kidney Dis.* 2014;63(3):491–499.
 230. Johansen KL, Chertow GM, Gilbertson DT, et al. US renal data system 2022 annual data report: Epidemiology of kidney disease in the United States. *Am J Kidney Dis.* 2023;81(3 Suppl1):A8–A11.
 231. de Rooij ENM, Meuleman Y, de Fijter JW, et al. Quality of life before and after the start of dialysis in older patients. *Clin J Am Soc Nephrol.* 2022;17(8):1159–1167.

232. de Rooij ENM, Meuleman Y, de Fijter JW, et al. Symptom burden before and after dialysis initiation in older patients. *Clin J Am Soc Nephrol*. 2022;17(12):1719–1729.
233. Fonseca-Correa JI, Jassal SV. Health care for older adults with kidney failure. *Clin J Am Soc Nephrol*. 2022;17(8):1110–1112.
234. Johansen KL, Chertow GM, Gilbertson DT, et al. US renal data system 2021 annual data report: Epidemiology of kidney disease in the United States. *Am J Kidney Dis*. 2022;79(4 Suppl 1):A8–A12.
235. Cohen LM, Ruthazer R, Moss AH, Germain MJ. Predicting six-month mortality for patients who are on maintenance hemodialysis. *Clin J Am Soc Nephrol*. 2010;5(1):72–79.
236. Floege J, Gillespie IA, Kronenberg F, et al. Development and validation of a predictive mortality risk score from a European hemodialysis cohort. *Kidney Int*. 2015;87(5):996–1008.
237. Kan WC, Wang JJ, Wang SY, et al. The new comorbidity index for predicting survival in elderly dialysis patients: a long-term population-based study. *PLoS One*. 2013;8(8):e68748.
238. Ng YY, Hung YN, Wu SC, Ko PJ, Hwang SM. Progression in comorbidity before hemodialysis initiation is a valuable predictor of survival in incident patients. *Nephrol Dial Transplant*. 2013;28(4):1005–1012.
239. Brar A, Mallappallil M, Stefanov DG, Kau D, Salifu MO. Dialysis in the elderly and impact of institutionalization in the United States renal data system. *Am J Nephrol*. 2017;46(2):114–119.
240. Mauri JM, Cleries M, Vela E, Catalan Renal R. Design and validation of a model to predict early mortality in haemodialysis patients. *Nephrol Dial Transplant*. 2008;23(5):1690–1696.
241. Schmidt RJ, Landry DL, Cohen L, et al. Derivation and validation of a prognostic model to predict mortality in patients with advanced chronic kidney disease. *Nephrol Dial Transplant*. 2019;34(9):1517–1525.
242. Couchoud C, Hemmelgarn B, Kotanko P, Germain MJ, Moranne O, Davison SN. Supportive care: time to change our prognostic tools and their use in CKD. *Clin J Am Soc Nephrol*. 2016;11(10):1892–1901.
243. Han SS, Park JY, Kang S, et al. Dialysis modality and mortality in the elderly: a meta-analysis. *Clin J Am Soc Nephrol*. 2015;10(6):983–993.
244. Brown EA, Finkelstein FO, Iyasere OU, Kliger AS. Peritoneal or hemodialysis for the frail elderly patient, the choice of 2 evils? *Kidney Int*. 2017;91(2):294–303.
245. Jassal SV, Watson D. Offering peritoneal dialysis to the older patient: medical progress or waste of time? *Semin Nephrol*. 2011;31(2):225–234.
246. Oliver MJ, Quinn RR. Selecting peritoneal dialysis in the older dialysis population. *Perit Dial Int*. 2015;35(6):618–621.
247. Farragher JF, Oliver MJ, Jain AK, Flanagan S, Koyle K, Jassal SV. PD assistance and relationship to Co-existing geriatric syndromes in incident peritoneal dialysis therapy patients. *Perit Dial Int*. 2019;39(4):375–381.
248. Wong SN, Turnbull L, Saunders S, et al. Evaluation and outcomes of a 5-year assisted peritoneal dialysis program. *Perit Dial Int*. 2023;8968608221149546.
249. Fonseca-Correa JI, Farragher JF, Tomlinson G, et al. Longitudinal changes in the use of PD assistance for patients maintained on peritoneal dialysis. *Kidney360*. 2021;2(3):469–476.
250. Brown EA. How to address barriers to peritoneal dialysis in the elderly. *Perit Dial Int*. 2011;31(Suppl 2):S83–S85.
251. Shea YF, Lam MF, Lee MS, et al. Prevalence of cognitive impairment among peritoneal dialysis patients, impact on peritonitis and role of assisted dialysis. *Perit Dial Int*. 2016;36(3):284–290.
252. Brown EA, Bargman JM, Li PK. Managing older patients on peritoneal dialysis. *Perit Dial Int*. 2015;35(6):609–611.
253. Dimkovic N, Oreopoulos D. Management of elderly patients with end-stage kidney disease. *Semin Nephrol*. 2009;29(6):643–649.
254. Findlay MD, Dawson J, Dickie DA, et al. Investigating the relationship between cerebral blood flow and cognitive function in hemodialysis patients. *J Am Soc Nephrol*. 2019;30(1):147–158.
255. Cabrera VJ, Hansson J, Kliger AS, Finkelstein FO. Symptom management of the patient with CKD: the role of dialysis. *Clin J Am Soc Nephrol*. 2017;12(4):687–693.
256. Lentine KL, Smith JM, Miller JM, et al. OPTN/SRTR 2021 annual data report: kidney. *Am J Transplant*. 2023;23(2 Suppl 1):S21–S120.
257. Hod T, Goldfarb-Rumyantsev AS. Clinical issues in renal transplantation in the elderly. *Clin Transplant*. 2015;29(2):167–175.
258. Danovitch G, Savransky E. Challenges in the counseling and management of older kidney transplant candidates. *Am J Kidney Dis*. 2006;47(4):S86–S97.
259. Thind AK, Rule A, Goodall D, et al. Prevalence of frailty and cognitive impairment in older transplant candidates - a preview to the Kidney Transplantation in Older People (KTOP): impact of frailty on outcomes study. *BMC Nephrol*. 2022;23(1):283.
260. Alfieri C, Malvica S, Cesari M, et al. Frailty in kidney transplantation: a review on its evaluation, variation and long-term impact. *Clin Kidney J*. 2022;15(11):2020–2026.
261. Pérez-Sáez MJ, Arias-Cabrales CE, Redondo-Pachón D, et al. Increased mortality after kidney transplantation in mildly frail recipients. *Clin Kidney J*. 2022;15(11):2089–2096.
262. Segall L, Nistor I, Pascual J, et al. Criteria for and appropriateness of renal transplantation in elderly patients with end-stage renal disease: a literature review and position statement on behalf of the European renal association-European dialysis and transplant association descartes working group and European renal best practice. *Transplantation*. 2016;100(10):e55–e65.
263. Wolfe RA, Ashby VB, Milford EL, et al. Comparison of mortality in all patients on dialysis, patients on dialysis awaiting transplantation, and recipients of a first cadaveric transplant. *N Engl J Med*. 1999;341(23):1725–1730.
264. Rao PS, Merion RM, Ashby VB, Port FK, Wolfe RA, Kayler LK. Renal transplantation in elderly patients older than 70 years of age: results from the Scientific Registry of Transplant Recipients. *Transplantation*. 2007;83(8):1069–1074.
265. Tso PL, Dar WA, Henry ML. With respect to elderly patients: finding kidneys in the context of new allocation concepts. *Am J Transplant*. 2012;12(5):1091–1098.
266. Molnar MZ, Streja E, Kovesdy CP, et al. Age and the associations of living donor and expanded criteria donor kidneys with kidney transplant outcomes. *Am J Kidney Dis*. 2012;59(6):841–848.
267. Doyle SE, Matas AJ, Gillingham K, Rosenberg ME. Predicting clinical outcome in the elderly renal transplant recipient. *Kidney Int*. 2000;57(5):2144–2150.
268. Schratzberger G, Mayer G. Age and renal transplantation: an interim analysis. *Nephrol Dial Transplant*. 2003;18(3):471–476.
269. Meier-Kriesche HU, Ojo AO, Hanson JA, Kaplan B. Exponentially increased risk of infectious death in older renal transplant recipients. *Kidney Int*. 2001;59(4):1539–1543.
270. Hakim FT, Flomerfelt FA, Boyiadzis M, Gress RE. Aging, immunity and cancer. *Curr Opin Immunol*. 2004;16(2):151–156.
271. Ellam T, El-Kossi M, Prasanth KC, El-Nahas M, Khwaja A. Conservatively managed patients with stage 5 chronic kidney disease—outcomes from a single center experience. *QJM*. 2009;102(8):547–554.
272. Joly D, Anglicheau D, Alberti C, et al. Octogenarians reaching end-stage renal disease: cohort study of decision-making and clinical outcomes. *J Am Soc Nephrol*. 2003;14(4):1012–1021.
273. Smith C, Da Silva-Gane M, Chandna S, Warwicker P, Greenwood R, Farrington K. Choosing not to dialyse: evaluation of planned non-dialytic management in a cohort of patients with end-stage renal failure. *Nephron Clin Pract*. 2003;95(2):c40–c46.
274. Wong CF, McCarthy M, Howse ML, Williams PS. Factors affecting survival in advanced chronic kidney disease patients who choose not to receive dialysis. *Ren Fail*. 2007;29(6):653–659.
275. Murtagh FE, Marsh JE, Donohoe P, Ekbil NJ, Sheerin NS, Harris FE. Dialysis or not? A comparative survival study of patients over 75 years with chronic kidney disease stage 5. *Nephrol Dial Transplant*. 2007;22(7):1955–1962.
276. Brown MA, Collett GK, Josland EA, Foote C, Li Q, Brennan FP. CKD in elderly patients managed without dialysis: survival, symptoms, and quality of life. *Clin J Am Soc Nephrol*. 2015;10(2):260–268.
277. Hemmelgarn BR, James MT, Manns BJ, et al. Rates of treated and untreated kidney failure in older vs younger adults. *JAMA*. 2012;307(23):2507–2515.
278. Wong SP, Hebert PL, Laundry RJ, et al. Decisions about renal replacement therapy in patients with advanced kidney disease in the US department of Veterans Affairs, 2000–2011. *Clin J Am Soc Nephrol*. 2016;11(10):1825–1833.

279. Hsu CY, McCulloch CE, Fan D, Ordonez JD, Chertow GM, Go AS. Community-based incidence of acute renal failure. *Kidney Int*. 2007;72(2):208–212.
280. Carson RC, Juszczak M, Davenport A, Burns A. Is maximum conservative management an equivalent treatment option to dialysis for elderly patients with significant comorbid disease? *Clin J Am Soc Nephrol*. 2009;4(10):1611–1619.
281. Wongrakpanich S, Susantitaphong P, Isaranuwatthai S, Chenbhanich J, Eiam-Ong S, Jaber BL. Dialysis therapy and conservative management of advanced chronic kidney disease in the elderly: a systematic review. *Nephron*. 2017;137(3):178–189.
282. Murtagh FE, Burns A, Moranne O, Morton RL, Naicker S. Supportive care: comprehensive conservative care in end-stage kidney disease. *Clin J Am Soc Nephrol*. 2016;11(10):1909–1914.
283. Davison SN, Jhangri GS. Impact of pain and symptom burden on the health-related quality of life of hemodialysis patients. *J Pain Symptom Manage*. 2010;39(3):477–485.
284. Davison SN, Koncicki H, Brennan F. Pain in chronic kidney disease: a scoping review. *Semin Dial*. 2014;27(2):188–204.
285. Wong SP, Kreuter W, O'Hare AM. Treatment intensity at the end of life in older adults receiving long-term dialysis. *Arch Intern Med*. 2012;172(8):661–663; discussion 3–4.
286. Group. ORw. *Rehabilitation program definitions Clinical Committees Survey Report: Current Status of Rehabilitation in the GTA*. 1999.
287. Wells JL, Seabrook JA, Stolee P, Borrie MJ, Knoefel F. State of the art in geriatric rehabilitation. Part I: review of frailty and comprehensive geriatric assessment. *Arch Phys Med Rehabil*. 2003;84(6):890–897.
288. Hiraki K, Yasuda T, Hotta C, et al. Decreased physical function in pre-dialysis patients with chronic kidney disease. *Clin Exp Nephrol*. 2013;17(2):225–231.
289. Johansen KL, Painter P. Exercise in individuals with CKD. *Am J Kidney Dis*. 2012;59(1):126–134.
290. Johansen KL, Chertow GM, Kutner NG, Dalrymple LS, Grimes BA, Kaysen GA. Low level of self-reported physical activity in ambulatory patients new to dialysis. *Kidney Int*. 2010;78(11):1164–1170.
291. McIntyre CW, Selby NM, Sigrist M, Pearce LE, Mercer TH, Naish PF. Patients receiving maintenance dialysis have more severe functionally significant skeletal muscle wasting than patients with dialysis-independent chronic kidney disease. *Nephrol Dial Transplant*. 2006;21(8):2210–2216.
292. Avesani CM, Trolonge S, Deleaval P, et al. Physical activity and energy expenditure in haemodialysis patients: an international survey. *Nephrol Dial Transplant*. 2012;27(6):2430–2434.
293. Gill TM, Allore HG, Holford TR, Guo Z. Hospitalization, restricted activity, and the development of disability among older persons. *JAMA*. 2004;292(17):2115–2124.
294. Kasbia GS, Farragher J, Kim SJ, Famure O, Jassal SV. A cross-sectional study examining the functional independence of elderly individuals with a functioning kidney transplant. *Transplantation*. 2014;98(8):864–870.
295. Li M, Porter E, Lam R, Jassal SV. Quality improvement through the introduction of interdisciplinary geriatric hemodialysis rehabilitation care. *Am J Kidney Dis*. 2007;50(1):90–97.
296. Farragher J, Einbinder Y, Oliver MJ, Nachman J, Chow E, Jassal SV. Importance of early inpatient geriatric rehabilitation on outcomes in individuals on dialysis. *Arch Phys Med Rehabil*. 2020;101(2):227–233.
297. Lo D, Chiu E, Jassal SV. A prospective pilot study to measure changes in functional status associated with hospitalization in elderly dialysis-dependent patients. *Am J Kidney Dis*. 2008;52(5):956–961.
298. Jassal SV, Chiu E, Li M. Geriatric hemodialysis rehabilitation care. *Adv Chronic Kidney Dis*. 2008;15(2):115–122.
299. Davison SN, Jassal SV. Supportive care: integration of patient-centered kidney care to manage symptoms and geriatric syndromes. *Clin J Am Soc Nephrol*. 2016;11(10):1882–1891.
300. Grubbs V, Moss AH, Cohen LM, et al. A palliative approach to dialysis care: a patient-centered transition to the end of life. *Clin J Am Soc Nephrol*. 2014;9(12):2203–2209.
301. O'Hare AM, Song MK, Kurella Tamura M, Moss AH. Research priorities for palliative care for older adults with advanced chronic kidney disease. *J Palliat Med*. 2017;20(5):453–460.
302. Romano TG, Palomba H. Palliative dialysis: a change of perspective. *J Clin Med Res*. 2014;6(4):234–238.
303. Vandecasteele SJ, Kurella Tamura M. A patient-centered vision of care for ESRD: dialysis as a bridging treatment or as a final destination? *J Am Soc Nephrol*. 2014;25(8):1647–1651.
304. Churchill DN, Jassal SV. Dialysis: destination or journey. *J Am Soc Nephrol*. 2014;25(8):1609–1611.
305. Davison SN. End-of-life care preferences and needs: perceptions of patients with chronic kidney disease. *Clin J Am Soc Nephrol*. 2010;5(2):195–204.
306. Cockcroft DW, Gault MH. Prediction of creatinine clearance from serum creatinine. *Nephron*. 1976;16(1):31–41.
307. Levey AS, Bosch JP, Lewis JB, Greene T, Rogers N, Roth D. A more accurate method to estimate glomerular filtration rate from serum creatinine: a new prediction equation. Modification of Diet in Renal Disease Study Group. *Ann Intern Med*. 1999;130(6):461–470.
308. Inker LA, Schmid CH, Tighiouart H, et al. Estimating glomerular filtration rate from serum creatinine and cystatin C. *N Engl J Med*. 2012;367(1):20–29.
309. Nyman U, Grubb A, Larsson A, et al. The revised Lund-Malmö GFR estimating equation outperforms MDRD and CKD-EPI across GFR, age and BMI intervals in a large Swedish population. *Clin Chem Lab Med*. 2014;52(6):815–824.
310. Pottel H, Delanaye P, Schaeffner E, et al. Estimating glomerular filtration rate for the full age spectrum from serum creatinine and cystatin C. *Nephrol Dial Transplant*. 2017;32(3):497–507.
311. Pottel H, Hoste L, Dubourg L, et al. An estimated glomerular filtration rate equation for the full age spectrum. *Nephrol Dial Transplant*. 2016;31(5):798–806.
312. Schaeffner ES, Ebert N, Delanaye P, et al. Two novel equations to estimate kidney function in persons aged 70 years or older. *Ann Intern Med*. 2012;157(7):471–481.
313. Rossier A, Pruijm M, Hannane D, Burnier M, Teta D. Incidence, complications and risk factors for severe falls in patients on maintenance haemodialysis. *Nephrol Dial Transplant*. 2012;27(1):352–357.
314. Kutner NG, Zhang R, Huang Y, Wasse H. Falls among hemodialysis patients: potential opportunities for prevention? *Clin Kidney J*. 2014;7(3):257–263.
315. Desmet C, Beguin C, Swine C, Jadoul M, Universite Catholique de Louvain Collaborative G. Falls in hemodialysis patients: prospective study of incidence, risk factors, and complications. *Am J Kidney Dis*. 2005;45(1):148–153.
316. Koren MJ, Blumen HM, Ayers EI, Verghese J, Abramowitz MK. Cognitive dysfunction and gait abnormalities in CKD. *Clin J Am Soc Nephrol*. 2021;16(5):694–704.
317. Angalakuditi MV, Gomes J, Coley KC. Impact of drug use and comorbidities on in-hospital falls in patients with chronic kidney disease. *Ann Pharmacother*. 2007;41(10):1638–1643.
318. Woolcott JC, Richardson KJ, Wiens MO, et al. Meta-analysis of the impact of 9 medication classes on falls in elderly persons. *Arch Intern Med*. 2009;169(21):1952–1960.
319. Anstey KJ, von Sanden C, Luszcz MA. An 8-year prospective study of the relationship between cognitive performance and falling in very old adults. *J Am Geriatr Soc*. 2006;54(8):1169–1176.
320. Tilling LM, Darawil K, Britton M. Falls as a complication of diabetes mellitus in older people. *J Diabetes Complications*. 2006;20(3):158–162.
321. Li M, Tomlinson G, Naglie G, Cook WL, Jassal SV. Geriatric comorbidities, such as falls, confer an independent mortality risk to elderly dialysis patients. *Nephrol Dial Transplant*. 2008;23(4):1396–1400.
322. Roberts R, Jeffrey C, Carlisle G, Brierley E. Prospective investigation of the incidence of falls, dizziness and syncope in haemodialysis patients. *Int Urol Nephrol*. 2007;39(1):275–279.
323. Eyler RF, Unruh ML, Quinn DK, Vilay AM. Psychotherapeutic agents in end-stage renal disease. *Semin Dial*. 2015;28(4):417–426.
324. Atalay H, Solak Y, Biyik M, et al. Sertraline treatment is associated with an improvement in depression and health-related quality of life in chronic peritoneal dialysis patients. *Int Urol Nephrol*. 2010;42(2):527–536.
325. Taraz M, Khatami MR, Dashti-Khavidaki S, et al. Sertraline decreases serum level of interleukin-6 (IL-6) in hemodialysis patients with depression: results of a randomized double-blind, placebo-controlled clinical trial. *Int Immunopharmacol*. 2013;17(3):917–923.

326. Koo JR, Yoon JY, Joo MH, et al. Treatment of depression and effect of antidepressant treatment on nutritional status in chronic hemodialysis patients. *Am J Med Sci.* 2005;329(1):1–5.
327. Hosseini SH, Espahbodi F, Mirzadeh Goudarzi SM. Citalopram versus psychological training for depression and anxiety symptoms in hemodialysis patients. *Iran J Kidney Dis.* 2012;6(6):446–451.
328. Yazici AE, Erdem P, Erdem A, et al. Efficacy and tolerability of escitalopram in depressed patients with end stage renal disease: an open placebo-controlled study. *Klin Psikofarmakol B.* 2012;22(1):23–30.
329. Levy NB, Blumenfield M, Beasley Jr CM, et al. Fluoxetine in depressed patients with renal failure and in depressed patients with normal kidney function. *Gen Hosp Psychiatry.* 1996;18(1):8–13.
330. Blumenfield M, Levy NB, Spinowitz B, et al. Fluoxetine in depressed patients on dialysis. *Int J Psychiatry Med.* 1997;27(1):71–80.